

Exact Resonant Identities Beyond the Diagonal

Ordered Hurwitz jets, coincident Stieltjes products,
harmonic powers, and a complete Cayley obstruction

Research continuation prepared with ChatGPT
for Vladimir Reshetnikov's ProveIt project

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Abstract

We continue the exact-identity programme of the ProveIt manuscript and its incoming research reports at a fixed repository snapshot. Four directions yield proved results. First, we determine the complete Laurent expansion through the finite part of the depth-four ordered Hurwitz zeta function along every transverse straight direction at $(1, 1, 1, 1)$. Three additional coordinates beyond the previously known ordering function suffice; we give convergent representations, exact shift laws, a differential relation, a three-direction reconstruction, and the cancellation locus of the displayed nonsymmetric terms. Second, a coordinate-completed cotangent–Hurwitz transform evaluates one arbitrary Stieltjes derivative times any finite product of coincident reflected polygamma factors. Its resonant subtraction retains the whole parameter-dependent amplitude. Consequences include cubic differences, higher reflected moments, and log-Gamma families involving zeta derivatives and harmonic numbers. Third, the Dirichlet series of raw harmonic powers with a continuously shifted outer denominator has polynomial Laurent coefficients and finite parts at every nonpositive integer. Centering the shift at $1/2$ removes every nonpositive even pole, and the resulting values belong to the ordinary multiple-zeta algebra without requiring Euler's constant. Fourth, an exact integer functional extends the existing bounded weight-seven obstruction to the entire convergent Cayley shuffle ideal plus the specified additional finite relation family. This constrains a proof strategy for S_6 ; it does not decide the period conjecture.

We provide complete analytic and algebraic arguments, a corrected sign in a specifically identified harmonic-power preprint formula, explicit integration guidance, and reproducible exact and numerical checks. The numerical diagnostics are distinguished throughout from proofs and interval certificates. The short S_6 , revised S_8 , generic cubic Tornheim reduction, and arithmetic minimality questions remain open.

Keywords. Multiple polylogarithm; ordered Hurwitz zeta; generalized Stieltjes constant; Hadamard finite part; polygamma reflection; harmonic-power Dirichlet series; Euler sums; shuffle algebra.

Contents

1	Scope, antecedents, and the exact meaning of the results	3
1.1	A ledger of contributions	3
1.2	Three normalizations that must not be interchanged	4
1.3	Reading and integration map	4
2	Independent directions at depth four	5
2.1	Conventions and the holomorphic regular germs	5
2.2	The small jet that controls the answer	6
2.3	Every transverse depth-four Laurent constant	7
2.4	Three positive rays recover all three coordinates	9
2.5	Absolutely convergent representatives	9
2.6	Exact shift and differential compatibility	10
3	Coincident reflected Stieltjes products close in ordinary zeta jets	11
3.1	A classical meromorphic transform and its local completion	11
3.2	Polygamma specializations and exact finite endpoint anomalies	13
3.3	Arbitrarily many coincident reflected factors	14
3.4	Log-Gamma primitives and a harmonic-number family	15
3.5	Extension to repeated seams at arbitrary shifts	16
3.6	Verification and remaining questions	17
4	Raw harmonic powers: polynomial finite parts and a centered cancellation	17
4.1	Every principal part has a finite Bernoulli formula	18
4.2	A finite polylogarithm calculation determines every finite part	19
4.3	The complete expansion through the finite part at zero	21
4.4	Explicit formulas at the next two spectral integers	22
4.5	Exact shift derivatives and normalized antiderivatives	23
5	The complete Cayley ideal does not close the bounded weight-seven search	24
5.1	The target and the exact formal presentation	24
5.2	Why the depth-preserving projector controls the entire ideal	26
5.3	The integer certificate and the proof	28
5.4	Consequences for further research	29
6	Corrections, source audit, and integration safeguards	29
6.1	What was inspected and what was not established	29
6.2	A sign correction and its full logarithmic extension	29
6.3	Five safeguards with mathematical consequences	30
6.4	Proposed canonical status updates	31
7	Further research questions and a programme of exact identities	31
7.1	Ordered jets, derivatives, and antiderivatives	32
7.2	Coincident products and Gamma primitives	32
7.3	Harmonic powers and spectral regular coefficients	33
7.4	The Gaussian period candidates	33
8	Proof dependencies and reproducible verification	34
8.1	Exact calculations	34
8.2	Independent numerical diagnostics	35
8.3	Files and replay commands	35
	References	36

1 Scope, antecedents, and the exact meaning of the results

The starting point is the canonical ProveIt manuscript [1] and the eleven archives present in `docs/incoming` at repository commit

5a790187c8e186e41e2b990b4941cb7a1a3c7b6b.

The commit timestamp is 11 October 2026, 00:09:09 UTC. The incoming reports retain their 10 October dates. The delivered source manifest records the retrieved paths and hashes. This article is an integration candidate; it does not modify the canonical manuscript or promote unreviewed incoming statements by editorial fiat.

The immediate motivation is three explicit unfinished parts of the latest exact-identity programme. The ordered-Hurwitz report [2] asks for the first independent-slope formula beyond depth three and for the directions where its additional ordering coefficients disappear. The triple-Stieltjes report [3] provides finite closure for reflected factors with distinct seams, while coincidences require a separate normalization. The Gaussian depth projector report [13] proposes completing the Cayley ideal before repeating a finite reduction search. Sections 2, 3, and 5 answer these questions in the precise senses stated below. The harmonic-power work in Section 4 gives a further exact family connecting multiple polylogarithms, Gamma coefficients, harmonic numbers, and spectral finite parts.

1.1 A ledger of contributions

Direction	Proved here	Boundary of the claim
Depth-four ordered resonance	A spanning normal form for all transverse straight rays, convergent coordinates, shifts, and reconstruction	No arithmetic independence; no prescription for a ray contained in a polar hyperplane.
Coincident reflected products	All Stieltjes and argument derivative indices; arbitrary finite reflected products; explicit Gamma primitives	The generic unreflected cubic Tornheim coordinate is not evaluated.
Raw harmonic powers	All nonpositive principal and finite coefficients as shift polynomials; centered cancellation; explicit families	The first regular spectral coefficient generally needs global Mellin data.
Gaussian formal obstruction	Nonmembership after adding the complete Cayley ideal to the frozen additional relation span	S_6 and revised S_8 remain conjectural as numerical period identities.
Literature correction	A sign correction to equation (22) of arXiv:2507.03058v1, with an all-logarithmic proof	It is a correction to that identified formula, not a rejection of the preprint’s entire framework.

The word “new” in this report means an explicitly demonstrated continuation beyond the inspected ProveIt results. We identify the classical inputs and the nearest primary-source antecedents. We do not claim an exhaustive priority search over every formulation in the literature. In particular, the Gamma diagonal of ordered regular germs, ordinary Hurwitz Fourier expansions, shuffle regularization, and raw-power meromorphic continuation are antecedents.

The analytic context is the Laurent theory of multiple zeta functions developed by Saha [4] and by Matsumoto, Onozuka, and Wakabayashi [5]. Hurwitz integral methods and their Gamma applications have substantial classical development, including Espinosa and Moll [6, 7]. Tin [9] studies raw harmonic-power Dirichlet series and their harmonic Stieltjes constants; Kargin, Dil, Cenkci, and Can [8] study related harmonic zeta constants. We retain their

strict/inclusive and inner/outer shift distinctions rather than silently identifying different series. The shuffle-polynomial and Eulerian-logarithm structure in the formal argument is classical [14, 15]; the new certificate concerns its application to the complete ideal intersection.

1.2 Three normalizations that must not be interchanged

For $a > 0$, throughout the article,

$$\zeta(1 + u, a) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(a) u^m, \quad \gamma_0(a) = -\psi(a).$$

A spectral derivative differentiates the order s of a zeta function; an argument derivative differentiates its positive shift a or its position x . A superscript (p) on γ_m or ψ in the reflected section always denotes an argument derivative. Bracketed superscripts $[j]$ on z_k in the ordered section denote spectral derivatives. The ordinary constants are $\gamma_m = \gamma_m(1)$. Generalized harmonic numbers are $H_n^{(r)} = \sum_{j=1}^n j^{-r}$, with $H_0^{(r)} = 0$.

There are three different finite-coefficient operations.

1. A *spectral finite part* is the constant Laurent coefficient in a specified complex parameter. The ray variable ε in Section 2 and the spectral variable $s + m$ in Section 4 are explicitly fixed.
2. A *coordinate finite-part integral* is the constant term of a cut-off integral in fixed local coordinates. On $(0, 1)$ those coordinates are x and $1 - x$. Powers and logarithms are expanded before the constant term is selected. This is the convention in Section 3.
3. A *formal shuffle relation* is an equality in a specified rational word algebra. Its truth and exact verification are independent of an unproved assertion that the period map is injective.

All ordinary integrals, convergent sums, analytic continuations, and finite parts are identified where they first occur. Analytic continuation of an integral and coefficientwise coordinate completion agree away from a resonance, but can differ by a finite Taylor correction at one. That correction is computed explicitly in the coincident theorem.

The multiple-polylogarithm convention used for the Gaussian target is

$$\text{Li}_{r,s}(u, v) = \sum_{n > m \geq 1} \frac{u^n v^m}{n^r m^s}.$$

The one-variable multiple polylogarithms in the harmonic-power kernel carry the argument only on the outer index. These conventions fix the word order, colors, and harmonic conversions; changing them changes intermediate coordinate formulas.

1.3 Reading and integration map

The four mathematical sections can be read independently after the normalizations above. The ordered section supplies all of its required lower-depth germs. The reflected section proves the completed transform before using polynomial reduction. The harmonic section gives both a finite general algorithm and closed low-weight examples. The formal section includes the algebraic argument connecting its finite exact checks to the unrestricted ideal. Section 6 distinguishes an actual correction from normalization safeguards, and Section 7 identifies concrete remaining targets.

For canonical integration, the ordered section naturally follows the latest ordered-Hurwitz contribution; the reflected section follows the coincident/contact calculus; the harmonic section belongs with harmonic polylogarithm and endpoint regularization material; and the complete

Cayley certificate belongs immediately after the existing bounded S_6 obstruction. The retained proof-status qualifications are part of each mathematical statement.

2 Independent directions at depth four

The most recent ordered-Hurwitz report asks for an explicit description of the first independent-slope case beyond depth three [2, Questions 1–2]. We give such a description. Three additional coordinates suffice beyond its previously defined ordering function $\eta(a)$. All three have absolutely convergent representatives. The resulting formula includes the complete principal part, every transverse straight direction, exact shift recurrences, and a finite reconstruction of the coordinates from three positive directions. The construction is a spanning normal form; it does not assert arithmetic independence of its coordinates.

2.1 Conventions and the holomorphic regular germs

Fix $a > 0$ and use real logarithms of $n + a$. Set

$$Z_d(s_1, \dots, s_d; a) = \sum_{n_1 > \dots > n_d \geq 0} \prod_{j=1}^d (n_j + a)^{-s_j}, \quad Z_0 = 1.$$

The initial domain is $\Re(s_1 + \dots + s_j) > j$ for every j . Write $u_j = s_j - 1$ and $P_k = u_1 + \dots + u_k$. The Laurent convention is

$$\zeta(1 + u, a) = \frac{1}{u} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j(a)}{j!} u^j. \quad (2.1)$$

Throughout this section,

$$g_j = \gamma_j(a), \quad g = g_0 = -\psi(a), \quad z_k^{[j]} = \partial_s^j \zeta(s, a) \Big|_{s=k}, \quad z_k = z_k^{[0]}.$$

Thus a prime on an ordinary zeta value denotes its order derivative, whereas $\eta'(a)$ below denotes its shift derivative.

For completeness, the regular germs can be constructed directly. Define

$$D(s, x) = \zeta(s, x + 1) - \frac{x^{1-s}}{s-1}, \quad (2.2)$$

$$E_d(\mathbf{u}; a) = \sum_{n_2 > \dots > n_d \geq 0} D(1 + u_1, n_2 + a) \prod_{j=2}^d (n_j + a)^{-1-u_j}. \quad (2.3)$$

Let $R_0 = 1$, $R_1(u; a) = \zeta(1 + u, a) - 1/u$, and recursively put

$$R_d(\mathbf{u}; a) = E_d(\mathbf{u}; a) + \frac{R_{d-1}(u_1 + u_2, u_3, \dots; a) - R_{d-1}(u_2, u_3, \dots; a)}{u_1}. \quad (2.4)$$

The divided difference takes its removable value when $u_1 = 0$.

Lemma 2.1 (Ordered decomposition and normalization). *The germs R_d are holomorphic on $\sum_j |u_j| < 1/2$, and*

$$Z_d(1 + \mathbf{u}; a) = \sum_{k=0}^d \frac{R_{d-k}(u_{k+1}, \dots, u_d; a)}{P_1 \dots P_k}. \quad (2.5)$$

Furthermore, if $B_d(a) = R_d(0, \dots, 0; a)$, then

$$\sum_{d \geq 0} B_d(a) y^d = \frac{\Gamma(a)}{\Gamma(a + y)}. \quad (2.6)$$

Proof. For $\Re s > 0$, the tail defect has the integral

$$D(s, x) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-xt} \left(\frac{1}{e^t - 1} - \frac{1}{t} \right) dt.$$

The bracket is bounded at zero. This proves removability at $s = 1$ and $D(s, x) = O(x^{-\Re s})$ on compact subsets. The summands in (2.3), after summing the inner finite indices, have a uniform bound of the form $C(1 + n_2)^{-2 + \sum |u_{ij}|} (1 + \log(1 + n_2))^{d-2}$. Normal convergence and the integral expression for a divided difference prove holomorphy. Summing first over $n_1 > n_2$ gives

$$Z_d(1 + \mathbf{u}; a) = \frac{1}{u_1} Z_{d-1}(1 + u_1 + u_2, 1 + u_3, \dots; a) + E_d(\mathbf{u}; a),$$

so induction proves (2.5). On the common diagonal, the elementary-symmetric product for Z_d implies, coefficientwise as a formal power series in y ,

$$\sum_{d \geq 0} R_d(t, \dots, t; a) y^d = \exp \left\{ y R_1(t; a) + \sum_{m \geq 2} \frac{(-1)^{m-1}}{m} \zeta(m + mt, a) y^m \right\}. \quad (2.7)$$

At $t = 0$, the logarithmic Taylor expansion of $\Gamma(a + y)$ gives (2.6). $\square$

This construction and the diagonal generating function are antecedents, not new claims of this article. For $a = 1$, the ordered Laurent framework is due to Saha [4]; general integer-point Laurent expansions are studied in [5]. The common-shift construction above is the one in [2]. In particular,

$$B_2 = \frac{g^2 - z_2}{2}, \quad B_3 = \frac{g^3 - 3gz_2 + 2z_3}{6}, \quad (2.8)$$

$$B_4 = \frac{g^4 - 6g^2z_2 + 3z_2^2 + 8gz_3 - 6z_4}{24}. \quad (2.9)$$

2.2 The small jet that controls the answer

Define

$$\eta(a) = [u]R_2(u, 0; a), \quad (2.10)$$

$$\rho(a) = [u^2]R_2(u, 0; a) - [v^2]R_2(0, v; a), \quad (2.11)$$

$$\kappa(a) = (\partial_1 - 2\partial_2 + \partial_3)R_3(0, 0, 0; a), \quad (2.12)$$

$$T(a) = (\partial_u - \partial_v)Z_2(2 + u, 1 + v; a)|_{u=v=0}. \quad (2.13)$$

The factorial convention in (2.11) matters: ρ is a difference of Taylor coefficients, not a difference of second derivatives. The last coordinate has the elementary convergent series

$$T(a) = \sum_{n > m \geq 0} \frac{\log(m + a) - \log(n + a)}{(n + a)^2(m + a)}. \quad (2.14)$$

Introduce the ordinary-data combinations

$$D_1 = -gg_1 - z_2^{[1]}, \quad (2.15)$$

$$E = \frac{gg_2 - z_2^{[2]}}{2}, \quad F = \frac{g_1^2 - z_2^{[2]}}{2}, \quad (2.16)$$

$$S = -g_1B_2 - gz_2^{[1]} + z_3^{[1]}, \quad (2.17)$$

$$V = g\eta + \frac{g_1(g^2 + z_2)}{2} + gz_2^{[1]} - T. \quad (2.18)$$

Only V retains a nonordinary coordinate; the letters D_1, E, F, S denote explicitly displayed finite expressions.

Theorem 2.2 (Complete required jet). *The quadratic Taylor polynomial of R_2 and the linear Taylor polynomial of R_3 are*

$$R_2(u, v; a) = B_2 + \eta u + (D_1 - \eta)v + \frac{E + \rho}{2}u^2 + Fuv + \frac{E - \rho}{2}v^2 + O((|u| + |v|)^3), \quad (2.19)$$

$$R_3(u, v, w; a) = B_3 + \frac{S}{3}(u + v + w) + \frac{\kappa}{6}(u - 2v + w) + \frac{V}{2}(u - w) + O((|u| + |v| + |w|)^2). \quad (2.20)$$

Proof. The regularized depth-two stuffle identity is

$$R_2(u, v) + R_2(v, u) = R_1(u)R_1(v) - \zeta(2 + u + v, a). \quad (2.21)$$

It follows directly by substituting (2.5) into the ordinary stuffle identity; the rational polar terms cancel. Its coefficients of u , u^2 and uv , together with the definition of ρ , prove (2.19).

For the next depth put

$$C(u, v) = Z_2(2 + u, 1 + v; a), \quad L(u, v) = Z_2(1 + u, 2 + v; a) - \frac{\zeta(2 + v, a)}{u}.$$

The first function converges normally near zero. The second is holomorphic there because the two-factor stuffle law gives the exact expression

$$L(u, v) = R_1(u)\zeta(2 + v, a) - C(v, u) - \zeta(3 + u + v, a). \quad (2.22)$$

Applying the three-factor stuffle identity and cancelling the explicit prefix poles gives

$$R_1(u)R_2(v, w) = R_3(u, v, w) + R_3(v, u, w) + R_3(v, w, u) + C(u + v, w) + L(v, u + w). \quad (2.23)$$

For clarity, the needed cancellation uses $R_2(u, w) + R_2(w, u) = R_1(u)R_1(w) - \zeta(2 + u + w, a)$. Thus (2.23) follows from meromorphic identities and entails no unproved assumption that arbitrary finite-part prescriptions preserve products.

Write $X = \partial_1 R_3(0)$, $Y = \partial_2 R_3(0)$ and $Z = \partial_3 R_3(0)$. Differentiating (2.23) with respect to u and using (2.22) gives

$$X + Y + Z = -g_1 B_2 - g z_2^{[1]} + z_3^{[1]} = S.$$

Differentiating with respect to v instead gives

$$2X + Y = g\eta + g_1 z_2 + z_3^{[1]} - T.$$

Subtracting the preceding equation proves $X - Z = V$. Finally $X - 2Y + Z = \kappa$ by definition. Solving these three linear equations gives (2.20). $\square$

2.3 Every transverse depth-four Laurent constant

For $p, b, c, d \in \mathbb{C}$ suppose

$$p(p + b)(p + b + c)(p + b + c + d) \neq 0. \quad (2.24)$$

Here $\text{FP}_{\varepsilon=0}$ means the constant coefficient of the meromorphic function after the specified substitution. It is not an iterated residue and is not a value on a polar hyperplane.

Theorem 2.3 (Depth-four directional normal form). *Under (2.24),*

$$\begin{aligned} & \text{FP}_{\varepsilon=0} Z_4(1 + p\varepsilon, 1 + b\varepsilon, 1 + c\varepsilon, 1 + d\varepsilon; a) \\ &= B_4 + \frac{1}{p} \left\{ \frac{S(b+c+d)}{3} + \frac{\kappa(b-2c+d)}{6} + \frac{V(b-d)}{2} \right\} \\ & \quad + \frac{E(c^2+d^2)/2 + Fcd + \rho(c^2-d^2)/2}{p(p+b)} - \frac{g_3 d^3}{6p(p+b)(p+b+c)}. \end{aligned} \quad (2.25)$$

The complete principal part is

$$\begin{aligned} & \frac{1}{p(p+b)(p+b+c)(p+b+c+d)\varepsilon^4} + \frac{g}{p(p+b)(p+b+c)\varepsilon^3} \\ & + \frac{1}{\varepsilon^2} \left\{ \frac{B_2}{p(p+b)} - \frac{dg_1}{p(p+b)(p+b+c)} \right\} \\ & + \frac{1}{\varepsilon} \left\{ \frac{B_3}{p} + \frac{\eta(c-d) + D_1 d}{p(p+b)} + \frac{g_2 d^2}{2p(p+b)(p+b+c)} \right\}. \end{aligned} \quad (2.26)$$

Proof. At depth four, (2.5) has five terms:

$$\frac{1}{P_1 P_2 P_3 P_4} + \frac{R_1(u_4)}{P_1 P_2 P_3} + \frac{R_2(u_3, u_4)}{P_1 P_2} + \frac{R_3(u_2, u_3, u_4)}{P_1} + R_4(\mathbf{u}).$$

The constant term requires the cubic coefficient of R_1 , the quadratic coefficient of R_2 , the linear coefficient of R_3 , and $R_4(0) = B_4$. Equations (2.1), (2.19) and (2.20) give exactly (2.25). The lower powers give (2.26). Holomorphy of the remainders makes the coefficient calculation legitimate, including directions with some zero or negative slopes, provided the prefix sums remain nonzero. $\square$

The answer depends on p and the three trailing slopes, but it needs no new fourth-depth regular constant. The new information lies in lower-depth derivatives, exactly as the ordered pole geometry predicts.

Corollary 2.4 (Exact cancellation locus in this normal form). *Treat ρ, κ, T, η as named coordinates, without assigning unproved relations among their values. Under (2.24), all their coefficients in (2.25) vanish if and only if $b = c = d$. On this locus the answer reduces to ordinary Hurwitz jets.*

Proof. The coefficient of T is $-(b-d)/(2p)$ and the coefficient of κ is $(b-2c+d)/(6p)$. Their simultaneous vanishing forces $b = d$ and then $c = b$. This also kills the ρ and η coefficients. Conversely, substitution proves cancellation. $\square$

This corollary classifies cancellation in the proved coordinate expression. It does not say that no other direction can simplify after a new functional identity is discovered. In particular, it is not a statement of arithmetic independence at an individual value of a .

For $b = c = d = q$, the formula becomes

$$B_4 + \frac{q}{p} S + \frac{q^2}{p(p+q)} \left\{ \frac{g_1^2 + gg_2}{2} - z_2^{[2]} \right\} - \frac{q^3 g_3}{6p(p+q)(p+2q)}, \quad (2.27)$$

which recovers the already proved distinguished-slope formula of [2]. This is a consistency specialization, not an additional new evaluation.

2.4 Three positive rays recover all three coordinates

Define the ordinary part of (2.25) by

$$\mathcal{O}(p, b, c, d; a) = B_4 + \frac{S(b+c+d)}{3p} + \frac{E(c^2+d^2)/2 + Fcd}{p(p+b)} - \frac{g_3 d^3}{6p(p+b)(p+b+c)}. \quad (2.28)$$

Let $\Delta_A, \Delta_B, \Delta_C$ be the directional finite parts minus $\mathcal{O}$ at the respective rays

$$A = (1, 1, 2, 1), \quad B = (1, 1, 3, 1), \quad C = (1, 2, 1, 1).$$

All three lie in the positive convergence directions.

Corollary 2.5 (Finite reconstruction). *One has*

$$\rho = 2\Delta_B - 4\Delta_A, \quad (2.29)$$

$$\kappa = \frac{9}{2}\Delta_B - 12\Delta_A, \quad (2.30)$$

$$V = 2\Delta_C - \frac{3}{2}\Delta_B + 4\Delta_A. \quad (2.31)$$

Consequently T is recovered from (2.18) once η is known.

Proof. The three equations supplied by (2.25) are

$$\begin{pmatrix} \Delta_A \\ \Delta_B \\ \Delta_C \end{pmatrix} = \begin{pmatrix} -1/3 & 3/4 & 0 \\ -2/3 & 2 & 0 \\ 1/6 & 0 & 1/2 \end{pmatrix} \begin{pmatrix} \kappa \\ \rho \\ V \end{pmatrix}.$$

The determinant is $-1/12$. Inverting this rational matrix gives the displayed formulas. $\square$

Thus the appearance of three coordinates is observable by a finite exact linear calculation. The invertible matrix proves equivalence of two spanning descriptions; it does not prove that the three numerical values are independent over any field.

2.5 Absolutely convergent representatives

Put

$$Q_j(x) = \gamma_j(x+1) + \frac{\log^{j+1} x}{j+1}, \quad h_n(a) = \sum_{m=0}^{n-1} \frac{1}{m+a}, \quad \ell_n(a) = \sum_{m=0}^{n-1} \frac{\log(m+a)}{m+a}.$$

Here empty sums vanish. The familiar identities

$$h_n(a) = \psi(n+a) - \psi(a), \quad \ell_n(a) = \gamma_1(a) - \gamma_1(n+a)$$

make the representatives numerically usable with one outer summation.

Theorem 2.6 (Convergent coordinate formulas). *With $x = n+a$ in every sum,*

$$\eta(a) = \frac{g_2}{2} - \sum_{n \geq 0} \frac{Q_1(x)}{x}, \quad (2.32)$$

$$\rho(a) = \frac{g_3}{3} + \frac{1}{2} \sum_{n \geq 0} \frac{Q_2(x) - Q_0(x) \log^2 x}{x}, \quad (2.33)$$

$$\kappa(a) = \sum_{n \geq 0} \frac{-Q_1(x)h_n(a) + Q_0(x)(2 \log x h_n(a) - \ell_n(a))}{x} - \frac{3}{2}(E + \rho) + F. \quad (2.34)$$

All these series, as well as (2.14), converge absolutely and locally uniformly for a in compact subintervals of $(0, \infty)$.

Proof. Taylor expansion in (2.2) gives $D(1+u, x) = \sum_{j \geq 0} (-1)^j Q_j(x) u^j / j!$. If $r_{m_1, \dots, m_d}^{(d)} = [\mathbf{u}^{\mathbf{m}}] R_d$, the divided difference in (2.4) yields

$$r_{m_1, \dots, m_d}^{(d)} = [\mathbf{u}^{\mathbf{m}}] E_d + \binom{m_1 + m_2 + 1}{m_1 + 1} r_{m_1 + m_2 + 1, m_3, \dots, m_d}^{(d-1)}. \quad (2.35)$$

For $(m_1, m_2) = (1, 0), (2, 0), (0, 2)$ this gives the first two formulas. For $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, their combination with coefficients 1, $-2, 1$ gives the sum in (2.34) and the correction $-3[u^2]R_2 + [uv]R_2 = -3(E + \rho)/2 + F$. The estimates used for (2.3), followed by Cauchy's formula on a slightly larger compact spectral polydisc, justify every coefficient extraction and give absolute convergence. Alternatively, $Q_j(x) = O((1 + \log x)^j / x)$ suffices, since $h_n = O(1 + \log n)$ and $\ell_n = O(1 + \log^2 n)$. $\square$

These formulas define the coordinates independently of a numerical Laurent fit. They are useful even if no further ordinary-zeta reduction exists, and they keep the unevaluated part confined to explicitly convergent harmonic–Stieltjes sums.

2.6 Exact shift and differential compatibility

Proposition 2.7 (Triangular shift laws). *Let a superscript $+$ mean evaluation at $a+1$, and let $L = \log a$. Then*

$$\eta(a) - \eta(a+1) = -\frac{g_1^+}{a}, \quad (2.36)$$

$$\rho(a) - \rho(a+1) = \frac{g_2^+ - g^+ L^2}{2a}, \quad (2.37)$$

$$T(a) - T(a+1) = \frac{z_2^{[1],+} + L z_2^+}{a}, \quad (2.38)$$

$$\kappa(a) - \kappa(a+1) = \frac{3\eta^+ - 2D_1^+ - L B_2^+}{a}. \quad (2.39)$$

Proof. The terms with last index zero give the exact ordered shift identity

$$Z_d(\mathbf{s}; a) - Z_d(\mathbf{s}; a+1) = a^{-s_d} Z_{d-1}(s_1, \dots, s_{d-1}; a+1).$$

Substitution into (2.5) and induction on depth show that

$$R_d(\mathbf{u}; a) - R_d(\mathbf{u}; a+1) = a^{-1-u_d} R_{d-1}(u_1, \dots, u_{d-1}; a+1). \quad (2.40)$$

The first, second and fourth assertions are the indicated Taylor coefficients of (2.40). For the third, use $C(u, v; a) - C(u, v; a+1) = a^{-1-v} \zeta(2+u, a+1)$ and apply $\partial_u - \partial_v$. $\square$

Proposition 2.8 (A convergent derivative identity). *Let $C_0(a) = Z_2(2, 1; a)$. Then*

$$T(a) = -\eta'(a) - C_0(a) + g_1 z_2 + z_3^{[1]}. \quad (2.41)$$

At $a = 1$, the classical Euler evaluation $Z_2(2, 1; 1) = \zeta(3)$ gives

$$T(1) = -\eta'(1) - \zeta(3) + \gamma_1 \zeta(2) + \zeta'(3).$$

Proof. Differentiating the ordered defining sum in its initial domain, and then continuing, gives

$$\partial_a R_2(u, v; a) = -(1+u)C(u, v) - (1+v)L(u, v).$$

The pole removed from L cancels exactly with $\partial_a R_1(v)/u = -(1+v)\zeta(2+v, a)/u$. Apply ∂_u at zero and substitute $L_u(0, 0) = -g_1 z_2 - z_3^{[1]} - C_v(0, 0)$ from (2.22). The result is (2.41). Normal convergence justifies the derivatives of C ; the regular germs are holomorphic in a for $\Re a > 0$ on the principal branch. $\square$

Equations (2.36)–(2.41) connect the new directional coordinates to generalized harmonic sums, generalized Stieltjes constants, and parameter derivatives. They also supply independent consistency checks for implementations. A complete reduction of ρ or κ to ordinary single-zeta data is not proved here.

3 Coincident reflected Stieltjes products close in ordinary zeta jets

Let

$$K(x) = \pi \cot(\pi x), \quad \zeta(1+u, x) - \frac{1}{u} = \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(x) u^m.$$

In particular $\gamma_0(x) = -\psi(x)$. All finite parts in this section use *both* unit endpoint coordinates x and $1-x$. Equivalently,

$$\text{FP} \int_0^1 f(x) dx = \text{CT}_{\epsilon \downarrow 0} \int_\epsilon^{1-\epsilon} f(x) dx,$$

where the constant coefficient is taken after expanding every endpoint power and logarithm. Independent lower and upper cutoffs give the same answer because the two local contributions are added. This convention is invariant under $x \mapsto 1-x$, and it assigns zero to $\text{FP} \int_0^1 x^{-1} \log^j x dx$.

The incoming triple report proves finite closure for one Stieltjes factor [3] and any number of reflected polygamma factors whose seams are pairwise separated. The result below allows every reflected factor to collide with the Stieltjes seam. It is an exact closed subclass of the coincident product problem. It does not evaluate the unreflected cube $\text{FP} \int \psi^3$, whose remaining diagonal Tornheim third jet is a different coordinate. The Fourier transform and reflection inputs are classical; the contribution here is the full coordinate completion and its all-order coincident reflected consequences.

3.1 A classical meromorphic transform and its local completion

Lemma 3.1 (Cotangent–Hurwitz transform). *For $r \geq 0$, initially when $\Re s$ is sufficiently negative and then by meromorphic continuation,*

$$J_r(s) := \text{FP} \int_0^1 \zeta(s, x) K^{(r)}(x) dx = (s)_r \pi \cot \frac{\pi(s+r)}{2} \zeta(s+r). \quad (3.1)$$

At a spectral resonance the right side denotes the meromorphic continuation, not the coordinate finite part of a specialized coefficient.

Proof. The periodic principal value of K has nonzero Fourier coefficients $-\pi i \operatorname{sgn}(n)$ and zero mean. This follows, for example, by differentiating the Abel Fourier series for $\log(2 \sin \pi x)$ in distributions. The Hurwitz Fourier expansion [16, Eq. 25.11.9] gives, for $\Re s < 0$,

$$J_0(s) = 2\pi\Gamma(1-s)(2\pi)^{s-1} \cos(\pi s/2) \zeta(1-s).$$

The sum is absolutely convergent because its summands have magnitude $O(n^{\Re s-1})$. The Riemann functional equation [20] turns the displayed expression into $\pi \cot(\pi s/2) \zeta(s)$, with removable singularities interpreted by continuation.

For $\Re s < -r-1$, the singular term x^{-s} and sufficiently many of its derivatives vanish at zero. The smooth part $\zeta(s, 1+x)$ has the same periodic endpoint jets as $\zeta(s, x)$ at one. Consequently distributional integration by parts against $K^{(r)}$ agrees with the stated endpoint finite part and gives

$$J_r(s) = (-1)^r \text{FP} \int_0^1 \partial_x^r \zeta(s, x) K(x) dx = (s)_r J_0(s+r).$$

There are no unrecorded endpoint constants in this initial region: the smooth jets match and the remaining power vanishes. Finite endpoint Taylor subtraction supplies meromorphic continuation in s , proving (3.1) everywhere as a meromorphic identity. $\square$

Set $N = p + r$ and define $P_j(u) = (1 + u)_j$, so $P_0 = 1$. For $p, r \geq 0$ put

$$\mathcal{R}_{p,r}(u) = (-1)^p \left[P_N(u) \pi \cot \frac{\pi(N+1+u)}{2} \zeta(N+1+u) - \mathbf{1}_{N \text{ odd}} \frac{2N! \zeta(N+1)}{p!} \frac{P_p(u)}{u} \right]. \quad (3.2)$$

For $N = 0$ the first line is the removable expression $-\pi \tan(\pi u/2) \zeta(1+u)$.

Theorem 3.2 (All-index coincident reflected closure). *The function $\mathcal{R}_{p,r}$ is holomorphic near $u = 0$, and for every $m, p, r \geq 0$,*

$$\boxed{\text{FP} \int_0^1 \gamma_m^{(p)}(x) K^{(r)}(x) dx = (-1)^m m! [u^m] \mathcal{R}_{p,r}(u).} \quad (3.3)$$

Thus all these moments have a finite reduction to ordinary Riemann zeta jets at $N+1$ and generalized harmonic polynomials. In the exceptional case $N = 0$, ordinary Stieltjes constants occur instead.

Proof. Write $g(u, x) = \zeta(1+u, x) - 1/u$. The exact local splitting is

$$\partial_x^p g(u, x) = (-1)^p P_p(u) x^{-p-1-u} + h_p(u, x), \quad h_p(u, x) = \partial_x^p g(u, 1+x).$$

The function h_p is holomorphic jointly near $(u, x) = (0, 0)$. At the upper endpoint the entire factor $\partial_x^p g(u, 1-y)$ is holomorphic jointly near $(0, 0)$ in (u, y) . The cotangent expansion is

$$K^{(r)}(x) = (-1)^r r! x^{-r-1} - 2 \sum_{j \geq 1} \zeta(2j) \frac{(2j-1)!}{(2j-1-r)!} x^{2j-1-r},$$

where a term is zero when $2j-1 < r$. In the product of the varying singular power with this expansion, resonance with x^{-1-u} occurs exactly when $2j = N+1$, hence exactly when N is odd. Its complete coefficient is

$$A_{p,r}(u) = -2(-1)^p \frac{N!}{p!} \zeta(N+1) P_p(u).$$

The meromorphic integral of this local monomial on $(0, b)$ is $-A_{p,r}(u) b^{-u}/u$. Its coefficientwise coordinate finite-part generator is $-A_{p,r}(u)(b^{-u} - 1)/u$. Their difference is $-A_{p,r}(u)/u$; therefore the required completion is the meromorphic integral plus $A_{p,r}(u)/u$.

All other varying-power monomials have denominators nonzero at zero. Subtract finitely many local Taylor terms at both endpoints so that all remainders and any fixed finite number of parameter derivatives are bounded by $Cx^{-\eta}(1 + |\log x|^M)$, respectively $Cy^{-\eta}(1 + |\log y|^M)$, with $\eta < 1$. These integrals are normally convergent. The fixed integer endpoint powers from the smooth factors have analytic coefficients and their coordinate finite parts may be extracted coefficientwise. This proves holomorphy and identifies all Taylor coefficients.

The meromorphic integral itself is, by (3.1),

$$(-1)^p P_p(u) J_r(1+p+u) = (-1)^p P_N(u) \pi \cot \frac{\pi(N+1+u)}{2} \zeta(N+1+u).$$

The subtraction $-K^{(r)}/u$ when $p = 0$ contributes nothing because $K^{(r)}$ has coordinate finite-part mean zero. Adding the complete local correction just obtained gives exactly (3.2), and coefficient extraction in g proves (3.3). $\square$

Removing only the pole residue would replace $P_p(u)$ by $p!$ and lose finite terms. In particular, for odd N it would incorrectly omit the harmonic correction H_p in the next formula.

3.2 Polygamma specializations and exact finite endpoint anomalies

Let $H_0 = 0$. At Stieltjes index zero, (3.3) gives

$$\text{FP} \int_0^1 \psi^{(p)}(x) K^{(r)}(x) dx = \begin{cases} \pi^2/2, & N = 0, \\ 0, & N > 0 \text{ even}, \\ 2(-1)^{p+1} N! [\zeta'(N+1) + (H_N - H_p) \zeta(N+1)], & N \text{ odd}. \end{cases} \quad (3.4)$$

The first examples are

$$\begin{aligned} \text{FP} \int_0^1 \psi(x) K'(x) dx &= -2\zeta'(2) - 2\zeta(2), \\ \text{FP} \int_0^1 \psi'(x) K(x) dx &= 2\zeta'(2), \\ \text{FP} \int_0^1 \psi(x) K'''(x) dx &= -12\zeta'(4) - 22\zeta(4), \\ \text{FP} \int_0^1 \psi'(x) K''(x) dx &= 12\zeta'(4) + 10\zeta(4), \\ \text{FP} \int_0^1 \psi''(x) K'(x) dx &= -12\zeta'(4) - 4\zeta(4), \\ \text{FP} \int_0^1 \psi'''(x) K(x) dx &= 12\zeta'(4). \end{aligned}$$

These are consistent with finite-part integration by parts, including its boundary constants. For example,

$$\text{FP} \int_0^1 (\psi K)' dx = -2\zeta(2).$$

Indeed the constant terms of $\psi(x)K(x)$ at zero and one are $3\zeta(2)$ and $\zeta(2)$, respectively. Thus setting the integral of this ordinary derivative to zero would be incorrect in the present coordinate convention.

At $p = r = 0$, every Stieltjes index has the particularly short formula

$$\begin{aligned} \text{FP} \int_0^1 \gamma_m(x) K(x) dx &= 4m! \sum_{k=1}^{\lfloor (m+1)/2 \rfloor} \frac{(1 - 2^{-2k}) \zeta(2k) \gamma_{m-2k+1}}{(m-2k+1)!} \\ &\quad - \mathbf{1}_{2|m} 4m! (1 - 2^{-m-2}) \zeta(m+2). \end{aligned} \quad (3.5)$$

For $m = 0, 1, 2$ this gives

$$-\frac{\pi^2}{2}, \quad \frac{\pi^2 \gamma}{2}, \quad \pi^2 \gamma_1 - \frac{\pi^4}{12}.$$

The base constant can also be recovered formally from the finite constant of the separated Stieltjes–cotangent transform in the incoming triple report. The local proof above establishes that this formal limit has the stated same-point coordinate normalization, and supplies all argument-derivative resonances.

For $N > 0$ even and $m = 1$ there is the simple additional family

$$\text{FP} \int_0^1 \gamma_1^{(p)}(x) K^{(r)}(x) dx = \frac{(-1)^p N! \pi^2}{2} \zeta(N+1).$$

All higher coefficients in (3.2) are finite because $P_j(u) = j! \prod_{a=1}^j (1 + u/a)$ is a polynomial.

3.3 Arbitrarily many coincident reflected factors

Define derivative polynomials

$$D_0(X) = X, \quad D_{r+1}(X) = -(X^2 + \pi^2)D'_r(X).$$

Then $K^{(r)}(x) = D_r(K(x))$, and D_r has degree $r+1$ and leading coefficient $(-1)^r r!$. Consequently every polynomial $P(X)$ has a unique finite decomposition

$$P(X) = C + \sum_{r=0}^{\deg P-1} c_r D_r(X). \quad (3.6)$$

The constant is explicitly

$$C = \frac{P(i\pi) + P(-i\pi)}{2},$$

since $D_r(\pm i\pi) = 0$ for $r \geq 1$ and $D_0(\pm i\pi) = \pm i\pi$. The remaining coefficients follow by descending elimination of the leading powers, an exact terminating procedure.

Corollary 3.3 (Finite coincident reflected algebra). *For arbitrary $m, p \geq 0$ and any polynomial P , the finite part*

$$\text{FP} \int_0^1 \gamma_m^{(p)}(x) P(K(x)) dx = \sum_r c_r (-1)^m m! [u^m] \mathcal{R}_{p,r}(u)$$

has a finite evaluation in ordinary zeta jets and Stieltjes constants. The same holds after replacing $P(K)$ by any finite product of arbitrary derivatives $K^{(r_j)}$. Equivalently, it holds for any finite product of the reflected factors

$$(-1)^{r_j} \psi^{(r_j)}(1-x) - \psi^{(r_j)}(x).$$

Proof. The reflection formula [17, Eq. 5.15.6] identifies each balanced polygamma expression with $K^{(r_j)}$. The derivative-polynomial recurrence converts their product to a polynomial in K . Apply (3.6) and Theorem 3.2. The constant term contributes zero because $\text{FP} \int_0^1 \gamma_m^{(p)}(x) dx = 0$. For $p = 0$ this follows by integrating the shifted Hurwitz family and completing its single resonance; for $p \geq 1$ it follows directly by taking endpoint constant terms of $\gamma_m^{(p-1)}$. The smooth endpoint jets coincide and the singular derivative of $x^{-1} \log^m x$ has no constant term. $\square$

For powers alone, write $K^q = C_q + \sum_r c_{q,r} K^{(r)}$. A useful recurrence is

$$C_0 = 1, \quad C_1 = 0, \quad c_{1,0} = 1, \quad C_q = -\pi^2 C_{q-2}, \quad c_{q,r} = -\frac{c_{q-1,r-1}}{q-1} - \pi^2 c_{q-2,r},$$

with absent coefficients zero. It follows from $(K^{q-1})' = -(q-1)(K^q + \pi^2 K^{q-2})$. In particular,

$$\begin{aligned} K^2 &= -K' - \pi^2, \\ K^3 &= \frac{1}{2} K'' - \pi^2 K, \\ K^4 &= -\frac{1}{6} K''' + \frac{4}{3} \pi^2 K' + \pi^4, \\ K^5 &= \frac{1}{24} K^{(4)} - \frac{5}{6} \pi^2 K'' + \pi^4 K. \end{aligned}$$

Since the coordinate finite-part mean of every $K^{(r)}$ is zero,

$$\text{FP} \int_0^1 K(x)^q dx = \pi^q \cos(q\pi/2). \quad (3.7)$$

The resulting mixed digamma examples include

$$\text{FP} \int_0^1 \psi(x) K(x)^2 dx = 2\zeta'(2) + 2\zeta(2), \quad (3.8)$$

$$\text{FP} \int_0^1 \psi(x) K(x)^4 dx = 2\zeta'(4) - \frac{8\pi^2}{3}\zeta'(2) - \frac{109}{3}\zeta(4). \quad (3.9)$$

There is an all-degree elementary family:

$$\boxed{\text{FP} \int_0^1 \psi(x) K(x)^{2h+1} dx = \frac{(-1)^h \pi^{2h+2}}{2} \quad (h \geq 0).} \quad (3.10)$$

Besides the polynomial proof, reflection gives a short independent argument: averaging the integral with its reflected copy yields $-\frac{1}{2} \text{FP} \int K^{2h+2}$, which is (3.10) by (3.7).

There is also an exactly reducible family with two unreflected factors:

$$\text{FP} \int_0^1 \psi(x)^2 K(x)^{2h+1} dx = -\text{FP} \int_0^1 \psi(x) K(x)^{2h+2} dx. \quad (3.11)$$

To prove it, reflect the left side and use $\psi(1-x) = \psi(x) + K(x)$. The reflected value is its negative minus twice the right-hand moment before its displayed minus sign, while the remaining pure odd power has zero finite-part mean. Thus every moment in (3.11) also has a finite ordinary-zeta-jet reduction by Corollary 3.3. At $h = 0$ it says

$$\text{FP} \int_0^1 \psi(x)^2 K(x) dx = -2\zeta'(2) - 2\zeta(2).$$

A direct cubic consequence, retaining the distinction from the open unreflected cube, is

$$\text{FP} \int_0^1 [\psi(x)^3 - \psi(x)^2 \psi(1-x)] dx = 2\zeta'(2) + 2\zeta(2). \quad (3.12)$$

Indeed expanding ψK^2 gives $\psi\psi(1-x)^2 - 2\psi^2\psi(1-x) + \psi^3$, and reflection identifies the finite parts of the first two mixed cubic monomials. This determines a difference of cubic moments and leaves their common unreflected Tornheim coordinate free.

3.4 Log-Gamma primitives and a harmonic-number family

The same completed transform supplies the integrated Gamma level. At $s = 0$, the only resonance for $\zeta(s, x)K(x)$ is x^{-1-s} . Thus

$$\pi \cot(\pi s/2) \zeta(s) + \frac{1}{s}$$

is the generating germ for the coordinate finite parts of the spectral jets $\zeta^{[j]}(0, x)K(x)$. Its linear coefficient is $\zeta''(0) + \zeta(2)/2$, because $\zeta(0) = -1/2$ and $\pi \cot(\pi s/2) = 2/s - \zeta(2)s + O(s^3)$. Lerch's identity [16, Eq. 25.11.18] $\zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$ and the zero mean of K give

$$\text{FP} \int_0^1 \log \Gamma(x) K(x) dx = \zeta''(0) + \frac{1}{2}\zeta(2). \quad (3.13)$$

The constant $1/s$ is independent of the parameter, apart from its pole, so this specialization has no omitted finite derivative of a resonant amplitude.

For $r \geq 1$, coordinate integration by parts gives the particularly simple primitive law

$$\text{FP} \int_0^1 \log \Gamma(x) K^{(r)}(x) dx = -\text{FP} \int_0^1 \psi(x) K^{(r-1)}(x) dx. \quad (3.14)$$

Here the boundary constant difference is zero, and this can be checked rather than assumed. For $r = 1$, the constant of $\log \Gamma(x)K(x)$ at each endpoint is $-\gamma$. For $r \geq 2$, use

$$\log \Gamma(x) = -\log x - \gamma x + \sum_{j \geq 2} \frac{(-1)^j \zeta(j)}{j} x^j, \quad \log \Gamma(1-y) = \gamma y + \sum_{j \geq 2} \frac{\zeta(j)}{j} y^j.$$

The constant at both endpoints of $\log \Gamma(x)K^{(r-1)}(x)$ is $-(r-1)!\zeta(r)/r$. Logarithmic terms are excluded by the same constant-term prescription. This proves (3.14) directly from the cutoff fundamental theorem. Combining it with (3.4) yields

$$\text{FP} \int_0^1 \log \Gamma(x) K^{(r)}(x) dx = \begin{cases} -\pi^2/2, & r = 1, \\ 2(r-1)![\zeta'(r) + H_{r-1}\zeta(r)], & r \geq 2 \text{ even}, \\ 0, & r \geq 3 \text{ odd}. \end{cases}$$

For any polynomial P with decomposition (3.6), its log-Gamma moment is therefore a finite sum of these evaluations and (3.13), with the constant contribution $C \log(2\pi)/2$. This is an explicit primitive counterpart of the coincident reflected algebra.

The even powers have an especially compact expression involving only ordinary harmonic numbers:

$$\boxed{\text{FP} \int_0^1 \log \Gamma(x) K(x)^{2h} dx = \frac{(-1)^h \pi^{2h}}{2} \left[\log(2\pi) - H_{2h} + \frac{1}{2} H_h \right] \quad (h \geq 0).} \quad (3.15)$$

Here is an independent proof of the normalization. Initially for $\Re a > 2h - 1$,

$$V_h(a) := \int_0^1 \sin(\pi x)^a K(x)^{2h} dx = \pi^{2h-1} B\left(\frac{a-2h+1}{2}, h + \frac{1}{2}\right).$$

At $a = 0$, all endpoint exponents are even integers; none is -1 . Thus this beta continuation is holomorphic there, and its value and first derivative equal the coordinate finite parts of K^{2h} and $K^{2h} \log \sin(\pi x)$. The beta identity [19], Gamma reflection, and the digamma recurrence [18] give

$$V_h(0) = (-1)^h \pi^{2h}, \quad V_h'(0) = (-1)^h \pi^{2h} \left[-\log 2 + H_{2h} - \frac{1}{2} H_h \right].$$

Finally average the log-Gamma integral with its reflected copy and use $\log \Gamma(x) + \log \Gamma(1-x) = \log \pi - \log \sin(\pi x)$. This proves (3.15) in the same unit coordinates. For example,

$$\text{FP} \int_0^1 \log \Gamma(x) K(x)^2 dx = \frac{\pi^2}{2} [1 - \log(2\pi)],$$

$$\text{FP} \int_0^1 \log \Gamma(x) K(x)^4 dx = \frac{\pi^4}{2} \left[\log(2\pi) - \frac{4}{3} \right].$$

The beta/reflection proof uses classical formulas; it is included as an explicit infinite family and as an independent check on the primitive normalization, without a claim of global priority.

3.5 Extension to repeated seams at arbitrary shifts

The same argument completes the separated theorem by rational partial fractions. Any finite product of $K^{(r_j)}(x - a_j)$ is rational in $X = e^{2\pi i x}$, bounded at $X = 0, \infty$, and has poles only at the finitely many $e^{2\pi i a_j}$. At each pole, the principal parts of $K^{(r)}(x - a)$ form a triangular basis. Subtracting them leaves a constant. Hence a unique finite representation exists as

$$C + \sum_a \sum_{r=0}^{M_a-1} c_{a,r} K^{(r)}(x - a),$$

where M_a is the pole order at that seam. Multiplication by the ordinary function $\gamma_m^{(p)}(x)$ and coordinate finite-part integration are linear operations on its local Laurent-log expansion, so this rational identity remains valid under precisely the stated normalization. The terms with $a = 0$ are evaluated by (3.3); the other terms use the already proved separated Stieltjes–cotangent kernel and its contact-corrected argument derivatives. No limit through colliding separated distributions is required.

For completeness, if $I_m(a) = \text{FP} \int_0^1 \gamma_m(x) K(x-a) dx$ at $0 < a < 1$, $N = p + r$, and $h_{m,p} = (-1)^m m! e_{m+1}(p)$ for $p \geq 1$ (zero for $p = 0$), the existing one-factor contact law [1, 3] gives

$$\text{FP} \int_0^1 \gamma_m^{(p)}(x) K^{(r)}(x-a) dx = (-1)^r [I_m^{(N)}(a) - h_{m,p} K^{(N)}(a)].$$

Here $e_j(p)$ is the elementary symmetric polynomial of degree j in $1, 1/2, \dots, 1/p$, and is zero for $j > p$. To check signs, use $F_{m,p} = D^p G_m + h_{m,p} \delta^{(p)}$ and $K^{(N)}(-a) = (-1)^{N+1} K^{(N)}(a)$. Thus arbitrary repetitions and coincidences within the reflected algebra still reduce to finitely many ordinary Hurwitz-zeta jets, with trigonometric coefficients at the remaining distinct shifts.

3.6 Verification and remaining questions

The accompanying verifier computes independent endpoint Laurent-log series from the ordinary Hurwitz shift relation and integrates them term by term on $(0, 1/5)$ and $(4/5, 1)$. On the central interval it uses ordinary high-precision quadrature of Stieltjes/polygamma and cotangent functions. It compares these values against finite coefficients of (3.2), and separately integrates powers K^q without applying the derivative-polynomial reduction. Exact symbolic assertions verify the power reduction recurrence. These are independent floating-point diagnostics, not interval enclosures; the proof above establishes the all-index result.

The natural next questions are to classify which products with two unreflected Stieltjes factors and a reflected polynomial still close without a Tornheim jet, to derive a conductor-compatible version of the repeated-seam partial fractions, and to formalize the local coefficient completion independently of distributional products. The present results do not imply an arithmetic evaluation of the generic cubic Tornheim derivative or either fixed-weight Gaussian candidate.

4 Raw harmonic powers: polynomial finite parts and a centered cancellation

The elementary-symmetric harmonic numerators of the height-one family do not equal raw powers. For example, its second numerator is $H_n^2 - H_n^{(2)}$, whereas the family studied here contains H_n^2 itself. Let

$$H_n = \sum_{j=1}^n \frac{1}{j}, \quad H_0 = 0, \quad F_p(s, a) = \sum_{n=0}^{\infty} \frac{H_n^p}{(n+a)^s}, \quad p \geq 1, \quad a > 0, \quad \Re s > 1. \quad (4.1)$$

The inner harmonic numbers are unshifted; only the outer denominator moves. All finite parts below mean the constant Laurent coefficient of this meromorphic continuation in the variable s . They are not an assignment of an ordinary value to the divergent defining series.

Dirichlet series of raw harmonic powers, their poles, and reductions to Euler sums have already been studied, notably by Tin [9]. The polylogarithm and shuffle ingredients are also classical [10]. The contribution developed here is the continuous outer-shift normalization, its polynomial finite-part law at every nonpositive integer, the exact cancellation at the centered shift, and explicit low-weight identities with normalized parameter primitives. No claim of priority is made for raw-power meromorphic continuation or for the classical Euler sums used in the reductions.

4.1 Every principal part has a finite Bernoulli formula

Write $B_j(a)$ for the Bernoulli polynomial, with $B_1(a) = a - 1/2$, and introduce the formal series

$$U(z, a) = - \sum_{j \geq 1} \frac{B_j(a)}{j} z^j. \quad (4.2)$$

Only finitely many coefficients are used; convergence of this formal asymptotic series is neither required nor asserted.

Theorem 4.1 (All poles and all principal coefficients). *The function $F_p(s, a)$ continues meromorphically to the entire s -plane, locally uniformly in $a > 0$. Its only possible poles are $1, 0, -1, -2, \dots$. At $s = 1$ its principal part is*

$$\sum_{r=0}^p \binom{p}{r} \frac{r! \gamma^{p-r}}{(s-1)^{r+1}}. \quad (4.3)$$

At $s = -m$, $m \geq 0$, put

$$F_p(-m + \varepsilon, a) = \sum_{h=1}^p \frac{R_{p,m,h}(a)}{\varepsilon^h} + C_{p,m}(a) + O(\varepsilon).$$

Then every principal coefficient is the finite polynomial

$$R_{p,m,h}(a) = \frac{p!}{(p-h+1)!} [z^{m+1}] (\gamma + U(z, a))^{p-h+1}, \quad 1 \leq h \leq p. \quad (4.4)$$

In particular,

$$R_{p,m,p}(a) = -\frac{p!}{m+1} B_{m+1}(a). \quad (4.5)$$

Every $R_{p,m,h}$ has degree at most $m+1$ in a .

Proof. Put $x = n + a$. The classical shifted digamma expansion, obtained for example from Euler–Maclaurin [11], is

$$H_n = \log x + \gamma - \sum_{j=1}^J \frac{B_j(a)}{j x^j} + O(x^{-J-1}). \quad (4.6)$$

It holds uniformly when a ranges over a compact positive interval; the finite differentiated versions have the same uniformity. Raising a finite truncation to the fixed power p gives

$$H_n^p = \sum_{j=0}^J \sum_{r=0}^p \binom{p}{r} [z^j] (\gamma + U(z, a))^{p-r} x^{-j} (\log x)^r + O(x^{-J-1} (1 + \log^p x)).$$

Subtract these finitely many terms in the defining Dirichlet series. The remainder is normally convergent in $\Re s > -J$ on compact subsets. The subtracted sums are the corresponding linear combination of $(-1)^r \partial_s^r \zeta(s + j, a)$. Arbitrarily large J therefore provides the meromorphic continuation and justifies every fixed parameter derivative.

The only principal term of $(-1)^r \partial_s^r \zeta(s + j, a)$ at $s = 1 - j$ is $r!/(s + j - 1)^{r+1}$. At $j = 0$ this proves (4.3). At $j = m + 1$, set $r = h - 1$ to obtain (4.4). The case $r = p$ has no positive-degree coefficient in z , so the pole order at a nonpositive integer is at most p . Formula (4.5) and the degree assertion follow directly from the Bernoulli-polynomial coefficients. $\square$

Corollary 4.2 (Cancellation at every nonpositive even integer). *For every $p \geq 1$, the centered function $F_p(s, 1/2)$ is holomorphic at $s = 0, -2, -4, \dots$. Its only poles are at $s = 1, -1, -3, -5, \dots$; their orders are exactly $p + 1$ at 1 and exactly p at each negative odd integer.*

Proof. All odd Bernoulli polynomials vanish at $1/2$, including $B_1(1/2)$. Thus $U(z, 1/2)$ contains only even powers of z and every coefficient in (4.4) vanishes when m is even. At $m = 2r - 1$, the top coefficient is $-p!B_{2r}(1/2)/(2r)$, which is nonzero because $B_{2r}(1/2) = (2^{1-2r} - 1)B_{2r} \neq 0$. The leading coefficient at $s = 1$ is $p!$ by (4.3). $\square$

The cancellation concerns the entire Laurent principal part, not only the residue. In the centered variable $x = n + 1/2$ it is the exact consequence of $H_n = \log x + \gamma + 1/(24x^2) - 7/(960x^4) + \dots$ having only even inverse powers after its logarithm.

4.2 A finite polylogarithm calculation determines every finite part

For a positive-index word $\mathbf{k} = (k_1, \dots, k_d)$, use

$$\mathrm{Li}_{\mathbf{k}}(z) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z^{n_1}}{n_1^{k_1} \dots n_d^{k_d}}.$$

Define the finite combination

$$A_p(t) = \sum_{\mathbf{k} \text{ a composition of } p} \frac{p!}{k_1! \dots k_d!} \mathrm{Li}_{\mathbf{k}}(e^{-t}). \quad (4.7)$$

Ordering the distinct indices in the product H_n^p proves the exact identity

$$(1 - e^{-t}) \sum_{n \geq 0} H_n^p e^{-nt} = A_p(t), \quad \Gamma(s) F_p(s, a) = \int_0^\infty t^{s-1} K_{p,a}(t) dt, \quad K_{p,a}(t) = \frac{e^{-at} A_p(t)}{1 - e^{-t}}. \quad (4.8)$$

The integral is initially ordinary for $\Re s > 1$. The first identity is a finite stuffle decomposition: an ordered equality block of size k_j contributes $n_j^{-k_j}$ and its multiplicity is the displayed multinomial coefficient.

For completeness, the local calculation has an explicit finite recursion. With $L_{\mathbf{k}}(t) = \mathrm{Li}_{\mathbf{k}}(e^{-t})$, set $L_\emptyset = 1$. Then

$$\frac{d}{dt} L_{k_1, \dots, k_d}(t) = \begin{cases} -L_{k_1-1, k_2, \dots, k_d}(t), & k_1 > 1, \\ -L_{k_2, \dots, k_d}(t)/(e^t - 1), & k_1 = 1. \end{cases} \quad (4.9)$$

For a convergent word $k_1 \geq 2$, the constant is $L_{\mathbf{k}}(0) = \zeta(\mathbf{k})$. For a leading-one word, the integration constant is fixed by the shuffle relations and $L_1(t) = -\log(1 - e^{-t})$, whose constant coefficient in powers of $\log t$ is zero. This is the usual shuffle regularization. It is a finite reduction: multiplying a word with one fewer initial 1 by Li_1 and shuffling its iterated integral word isolates the desired initial-one word with a nonzero integer multiplicity. All other terms have fewer initial 1's. Induction reduces its constant to convergent multiple zeta values.

At a specified Taylor order, (4.9) uses only rational operations and finite integrations of logarithmic monomials. Explicitly, for a polynomial P ,

$$\begin{aligned} \int t^{-1} P(\log t) dt &= \int P(\ell) d\ell|_{\ell=\log t}, \\ \int t^j P(\log t) dt &= t^{j+1} \sum_{q=0}^{\deg P} \frac{(-1)^q P^{(q)}(\log t)}{(j+1)^{q+1}}, \quad j \geq 0. \end{aligned} \quad (4.10)$$

This supplies an implementable calculation with fixed endpoint constants, rather than an unspecified asymptotic coefficient.

Theorem 4.3 (Polynomial finite parts at all nonpositive integers). *For fixed $p \geq 1$ and $m \geq 0$, compute the finite local expansion*

$$K_{p,a}(t) = \sum_{j=-1}^m t^j \sum_{r=0}^p c_{p,j,r}(a) (\log t)^r + O(t^{m+1} (1 + |\log t|^p)). \quad (4.11)$$

The coefficients are provided by (4.7)–(4.10); they are polynomials in a of degree at most $j + 1$ with coefficients in the algebra of convergent multiple zeta values. Define

$$\frac{1}{\Gamma(-m + \varepsilon)} = \sum_{\ell \geq 1} g_{m,\ell} \varepsilon^\ell. \quad (4.12)$$

Then the exact finite part is

$$C_{p,m}(a) = \sum_{r=0}^p (-1)^r r! c_{p,m,r}(a) g_{m,r+1}. \quad (4.13)$$

In particular $C_{p,m}$ is a polynomial in a of degree at most $m + 1$. Its coefficients lie in the algebra generated by Euler's constant and convergent multiple zeta values. At the centered regular points one has the stronger exact identity

$$F_p(-2r, 1/2) = (2r)! c_{p,2r,0}(1/2). \quad (4.14)$$

These centered values lie in the multiple-zeta algebra without requiring Euler's constant.

Proof. Equations (4.9) and (4.10), with the fixed shuffle constants, construct the expansion by induction on the weight. The ordinary convergent expansion of $1/(e^t - 1)$ has a simple pole at zero and radius 2π . Finite integration of its analytic remainders preserves a remainder bounded by a power of t times a fixed logarithmic polynomial. Consequently the displayed finite expansion is an actual local expansion with the stated remainder, locally uniformly in a ; there are no negative powers below t^{-1} . Multiplication by e^{-at} proves the degree bound.

Subtract (4.11) inside the integral over $(0, 1)$. The remainder integral, together with the integral over $(1, \infty)$, is holomorphic in $\Re s > -m - 1$. Its fixed spectral derivatives are obtained by inserting powers of $\log t$, by ordinary domination. Thus

$$\Gamma(s) F_p(s, a) = \sum_{j=-1}^m \sum_{r=0}^p \frac{(-1)^r r! c_{p,j,r}(a)}{(s+j)^{r+1}} + J_{p,m}(s, a), \quad (4.15)$$

where $J_{p,m}$ is holomorphic near $s = -m$. The zero of $1/\Gamma(s)$ at $s = -m$ kills the holomorphic remainder and all terms with $j \neq m$ when taking the constant Laurent coefficient. Only the finitely many $j = m$ terms remain, and their product with (4.12) is precisely (4.13).

For reference the reciprocal-Gamma coefficients are themselves finite Bell expressions, obtained from

$$\begin{aligned} \frac{1}{\Gamma(-m + \varepsilon)} &= (-1)^m m! \varepsilon \exp((\gamma - H_m) \varepsilon) \\ &\quad + \sum_{k \geq 2} \frac{(-1)^{k+1} \zeta(k) - H_m^{(k)}}{k} \varepsilon^k. \end{aligned} \quad (4.16)$$

The recurrence for Gamma proves this formula directly from the Taylor series of $1/\Gamma(1 + \varepsilon)$. At a centered regular point, the highest possible logarithmic coefficient in (4.15) would produce a nonzero pole. Descending triangularly, Corollary 4.2 therefore forces $c_{p,2r,j}(1/2) = 0$ for every $j \geq 1$. Since $g_{2r,1} = (2r)!$, this proves (4.14). $\square$

Where the global information returns. The theorem evaluates the principal part and constant term. It does not evaluate all spectral derivatives using the same finite data. Already the coefficient of ε receives the generally nonzero contribution $(-1)^m m! J_{p,m}(-m, a)$ from the holomorphic remainder in (4.15), together with the other regular terms. In particular, differentiating a formula valid only for a nonpositive integer does not determine the spectral derivative there. This distinguishes the present finite identities from the higher harmonic Stieltjes coefficients.

4.3 The complete expansion through the finite part at zero

Set

$$P_p(L) = p![u^p] \exp(Lu) \Gamma(1+u) e^{\gamma u}, \quad P_0 = 1. \quad (4.17)$$

Define ordinary convergent Euler sums

$$E_{q,k} = \sum_{n \geq 1} \frac{H_{n-1}^q}{n^k}, \quad q \geq 0, \quad k \geq 2. \quad (4.18)$$

At $q = 0$ the numerator is one. Every such sum is the finite multiple-zeta combination

$$E_{q,k} = \sum_{r \models q} \frac{q!}{r_1! \cdots r_d!} \zeta(k, r_1, \dots, r_d), \quad (4.19)$$

with the empty composition giving $E_{0,k} = \zeta(k)$.

Theorem 4.4 (An exact all-degree zero-resonance identity). *There are constants Q_p with $Q_1 = 1/2$ and the finite recurrence*

$$Q_p = -pQ_{p-1} + \frac{1}{2} \sum_{k=2}^{p-1} (k-1) \binom{p}{k+1} E_{p-1-k,k}, \quad p \geq 2. \quad (4.20)$$

For every $a > 0$ one has

$$\boxed{F_p(s, a) = \left(\frac{1}{2} - a\right) \frac{p! e^{\gamma s}}{s^p} + Q_p + O(s).} \quad (4.21)$$

Consequently

$$C_{p,0}(a) = Q_p + \left(\frac{1}{2} - a\right) \gamma^p, \quad F_p(0, 1/2) = Q_p. \quad (4.22)$$

Each Q_p has a finite expression using convergent multiple zeta values of weights at most $p-1$.

Proof. First the leading local polynomial of A_p is $P_p(-\log t)$. One way to fix it without a regularization guess is to use (4.3) in the Mellin identity. Multiplication by $\Gamma(1+s)$ turns the elementary pole polynomial at $s=0$ into exactly the coefficients of $\Gamma(1+u)e^{\gamma u}$ in (4.17). The word calculation proves existence of the local expansion, so this identifies its leading polynomial uniquely. At the centered shift $e^{-t/2}/(1-e^{-t}) = t^{-1} + O(t)$. Regularity at $s=0$ then forces the coefficient of t in A_p to have no logarithmic terms, by the same triangular argument as in the previous theorem. Hence

$$A_p(t) = P_p(-\log t) + tQ_p + O(t^2(1 + |\log t|^p)). \quad (4.23)$$

The elementary formula for $A_1 = -\log(1-e^{-t})$ gives $Q_1 = 1/2$.

To determine every other Q_p , put

$$M_{q,k}(t) = \sum_{n \geq 1} \frac{H_{n-1}^q}{n^k} e^{-nt}.$$

The exact increment $H_n^p - H_{n-1}^p$ gives

$$A_p = \sum_{k=1}^p \binom{p}{k} M_{p-k,k}, \quad A'_p = -\frac{pA_{p-1}}{e^t - 1} - \sum_{k=2}^p \binom{p}{k} M_{p-k,k-1}.$$

For $k \geq 2$, $M_{q,k}(t) \rightarrow E_{q,k}$ by domination. Using the preceding identity with $p-1$ also gives the complete leading logarithmic polynomial

$$M_{p-2,1}(t) = \frac{1}{p-1} \left(P_{p-1}(-\log t) - \sum_{k=2}^{p-1} \binom{p-1}{k} E_{p-1-k,k} \right) + o(1).$$

Take the coefficient of $t^0(\log t)^0$ in the derivative identity, using (4.23) and $(e^t - 1)^{-1} = t^{-1} - 1/2 + O(t)$. The two copies of P_{p-1} cancel as full polynomials. This leaves

$$Q_p = -pQ_{p-1} + \sum_{k=2}^{p-1} \left(\frac{p}{2} \binom{p-1}{k} - \binom{p}{k+1} \right) E_{p-1-k,k}.$$

The bracket is $(k-1)\binom{p}{k+1}/2$, proving the recurrence. The local expansions already established justify differentiation and coefficient comparison here.

It remains to show the particularly simple normalization in (4.21). The coefficient of t^0 in the kernel is $(1/2 - a)P_p(-\log t) + Q_p$. If $d(u) = \Gamma(1+u)e^{\gamma u}$, its P_p contribution to the Mellin continuation is

$$\frac{p!}{\Gamma(s)} [u^p] \frac{d(u)}{s-u}.$$

Replace $d(u)$ by $d(s)$: the difference quotient is holomorphic in (s, u) near $(0, 0)$ and its multiplication by $1/\Gamma(s)$ is $O(s)$. The remaining term is $p!e^{\gamma s}/s^p$. The constant Q_p contributes $Q_p/(s\Gamma(s)) = Q_p + O(s)$. All other local and global terms are $O(s)$ by the preceding theorem. This proves the result. $\square$

The first five constants are

$$\begin{aligned} Q_1 &= \frac{1}{2}, & Q_2 &= -1, \\ Q_3 &= 3 + \frac{1}{2}\zeta(2), & Q_4 &= -12 - 2\zeta(2) + 3\zeta(3), \\ Q_5 &= 60 + 10\zeta(2) - 15\zeta(3) + \frac{33}{2}\zeta(4). \end{aligned} \tag{4.24}$$

For example $E_{1,2} = \zeta(2, 1) = \zeta(3)$ and $E_{2,2} = 2\zeta(2, 1, 1) + \zeta(2, 2) = 11\zeta(4)/4$. These classical low-weight reductions follow from shuffle and stuffle relations. Equation (4.21) includes all negative Laurent powers: for example the cubic case is

$$F_3(s, a) = \left(\frac{1}{2} - a \right) \left(\frac{6}{s^3} + \frac{6\gamma}{s^2} + \frac{3\gamma^2}{s} + \gamma^3 \right) + 3 + \frac{1}{2}\zeta(2) + O(s).$$

4.4 Explicit formulas at the next two spectral integers

The finite word recursion gives the following local table. Here $L = -\log t$, $z_j = \zeta(j)$, and the omitted remainder is $O(t^4(1 + |\log t|^p))$.

$$\begin{aligned} A_1(t) &= L + \frac{t}{2} - \frac{t^2}{24} + O(t^4), \\ A_2(t) &= L^2 + z_2 - t - \frac{L}{12}t^2 - \frac{t^3}{36} + O(t^4(1 + L^2)), \\ A_3(t) &= L^3 + 3z_2L - 2z_3 + \left(3 + \frac{z_2}{2} \right) t \\ &\quad + \left(-\frac{L^2}{8} - \frac{L}{4} - \frac{z_2}{8} - \frac{3}{8} \right) t^2 + \frac{5}{72}t^3 + O(t^4(1 + |L|^3)), \end{aligned}$$

$$\begin{aligned}
A_4(t) = & L^4 + 6z_2L^2 - 8z_3L + \frac{27}{2}z_4 + (-12 - 2z_2 + 3z_3)t \\
& + \left(-\frac{L^3}{6} - \frac{L^2}{2} - \frac{z_2L}{2} - L - \frac{z_2}{2} + \frac{z_3}{3} - \frac{3}{4} \right) t^2 + \left(-\frac{z_2}{18} - \frac{7}{27} \right) t^3 + O(t^4(1 + |L|^4)).
\end{aligned} \tag{4.25}$$

This table is reproducible with (4.9). For clarity the only weight-four endpoint reductions required are

$$\zeta(3, 1) = \frac{1}{4}\zeta(4), \quad \zeta(2, 2) = \frac{3}{4}\zeta(4), \quad \zeta(2, 1, 1) = \zeta(4).$$

The leading-one regular constants follow by the indicated shuffle elimination; for example they are $-5\zeta(4)/4$ for $(1, 3)$, $-3\zeta(4)$ for $(1, 2, 1)$, and $3\zeta(4)$ for $(1, 1, 2)$. These constants, the two differentiation rules, and finite integration fully determine every entry of (4.25).

Corollary 4.5 (Uniform square, cube, and fourth-power identities). *For $1 \leq p \leq 4$, put $b = a - 1/2$ and*

$$T_1 = 0, \quad T_2 = -\frac{1}{12}, \quad T_3 = \frac{1}{8}, \quad T_4 = -\frac{1}{4}.$$

Then

$$F_p(-1 + \varepsilon, a) = p!e^{\gamma\varepsilon} \left(\frac{1/24 - b^2/2}{\varepsilon^p} + \frac{b^2}{2\varepsilon^{p-1}} \right) + bQ_p + T_p + O(\varepsilon). \tag{4.26}$$

In particular,

$$C_{p,1}(a) = \frac{\gamma^p}{24} + T_p + bQ_p + \frac{b^2}{2}(p\gamma^{p-1} - \gamma^p). \tag{4.27}$$

The centered regular values at $s = -2$ are

$$\begin{aligned}
F_1(-2, 1/2) &= -\frac{1}{24}, & F_2(-2, 1/2) &= \frac{1}{36}, \\
F_3(-2, 1/2) &= -\frac{1}{9} - \frac{\zeta(2)}{24}, & F_4(-2, 1/2) &= \frac{13}{27} + \frac{\zeta(2)}{18} - \frac{\zeta(3)}{4}.
\end{aligned} \tag{4.28}$$

Proof. Multiply (4.25) by $e^{-at}/(1 - e^{-t})$ and apply (4.13) at $m = 1$. Together with the Bernoulli formula for the principal part this gives (4.26). Equivalently, the centered calculation uses $e^{-t/2}/(1 - e^{-t}) = t^{-1} - t/24 + O(t^3)$ and the reciprocal-Gamma expansion at -1 ; the exact parameter transport proved next supplies its entire b -dependence. For $m = 2$, no logarithmic term survives at the centered shift, and (4.14) says to multiply the constant coefficient of t^2 by 2. This coefficient is $[t^3]A_p - Q_p/24$. Inserting the displayed table gives all four values in (4.28). $\square$

4.5 Exact shift derivatives and normalized antiderivatives

Let $R_{p,m}(a) = R_{p,m,1}(a)$ be the simple-pole coefficient in (4.4). It is given without a polylogarithm calculation by

$$R_{p,m}(a) = [z^{m+1}](\gamma + U(z, a))^p. \tag{4.29}$$

Theorem 4.6 (Parameter transport and finite primitives). *For $m \geq 1$,*

$$C'_{p,m}(a) = mC_{p,m-1}(a) - R_{p,m-1}(a). \tag{4.30}$$

At $m = 0$ the corresponding identity is $C'_{p,0}(a) = -\gamma^p$. For every $m \geq 0$ and positive a, a_0 one consequently has the normalized antiderivative identity

$$\int_{a_0}^a C_{p,m}(x) dx = \frac{C_{p,m+1}(a) - C_{p,m+1}(a_0) + \int_{a_0}^a R_{p,m}(x) dx}{m+1}. \tag{4.31}$$

The last integral is a finite polynomial integral explicitly specified by (4.29); it introduces no new special function or undetermined normalization.

Proof. Normal convergence in the defining half-plane gives $\partial_a F_p(s, a) = -s F_p(s+1, a)$, and the normally convergent continuations extend this identity meromorphically. Take the constant coefficient at $s = -m + \varepsilon$. The factor $m - \varepsilon$ contributes $m C_{p, m-1} - R_{p, m-1}$, proving (4.30). At $m = 0$, the residue at $s = 1$ is γ^p by (4.3). Integrating the relation with $m+1$ gives (4.31) and fixes its additive constant. $\square$

For example, substituting (4.22) and $R_{p,0}(a) = p(1/2 - a)\gamma^{p-1}$ into (4.30) proves the quadratic shift dependence (4.27) from its centered value. This provides an independent exact check on the a -polynomial obtained by Mellin coefficient extraction.

Extension and remaining questions. A polynomial in finitely many generalized harmonic numbers $H_n^{(r)}$ of positive integer orders r admits the same finite ordered-equality-block expansion. Replacing (4.7) by its finite multiple-polylogarithm combination (including a constant when the polynomial has one) therefore preserves the finite-part theorem, polynomial shift dependence, and normalized parameter transport. Its pole orders must be read from its own finite asymptotic expansion; the particular centered cancellation for raw powers does not automatically extend to arbitrary generalized-harmonic products. Three concrete questions remain: determine how far the Q_p reduce within specified multiple-zeta coordinate spaces at higher weight; produce compact all-degree formulas for the centered values $F_p(-2r, 1/2)$ beyond the word recursion; and identify combinations of first regular spectral coefficients whose global Mellin remainders cancel. The last question is distinct from the finite-part identities proved here.

5 The complete Cayley ideal does not close the bounded weight-seven search

The pinned manuscript leaves the mixed Gaussian identities for S_6 and S_8 conjectural. Two recent source contributions sharpened the algebraic problem in complementary ways. One supplied an exact integer functional separating the S_6 target from a specified collection of 5131 relations of depth at most four [12]. The other constructed a depth-preserving retraction for the *full* shuffle ideal generated by convergent Cayley identities [13]. It explicitly proposed projecting the bounded rows by that retraction as a next proof attempt.

We complete that proposed calculation. The inherited integer functional also annihilates the intersection of the *entire* Cayley shuffle ideal with the weight-seven space of depth at most four. This includes all Cayley rows and all their shuffle multiples, even when their construction uses higher-depth intermediate words. Thus completing the Cayley side alone cannot prove S_6 from the stated bounded additional relations. The numerical S_6 identity remains open. This is a result about an explicit formal presentation; it is not nonvanishing of its evaluated period residual, and it gives no period-independence assertion.

5.1 The target and the exact formal presentation

Use the convention

$$\mathrm{Li}_{r,s}(u, v) = \sum_{n > m \geq 1} \frac{u^n v^m}{n^r m^s}, \quad g_{r,s} = \mathrm{Im} \, \mathrm{Li}_{r,s}(i, 1), \quad S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}.$$

Let $L = \log 2$, and write the conjectural residual in its already reconciled short coordinate form:

$$R_6 = S_6 + \frac{12334}{527}g_{6,1} + \frac{384}{31}g_{5,2} + \frac{2136}{527}g_{4,3} - \frac{3179\pi^7}{5713920} + \frac{801}{2108}\beta(4)\zeta(3) + 2\beta(6)L. \quad (5.1)$$

The conjecture is $R_6 = 0$.

For formal words take the positive forms

$$z = \frac{dt}{t}, \quad c = \frac{dt}{1-t}, \quad a = \frac{idt}{1-it}, \quad b = -\frac{dt}{1+t}, \quad \bar{a} = -\frac{idt}{1+it}.$$

Their iterated integrals $H(w)$ are written outer letter first on $[0, 1]$. An admissible nonempty word begins with a letter other than c and ends with a letter other than z . Let $\mathcal{A}$ be the graded rational shuffle algebra on the admissible words and the empty word. Conjugation K interchanges $a, \bar{a}$ and fixes z, c, b . The superscript $-$ denotes its negative eigenspace, and $F_D\mathcal{A}$ is the span of words with at most D nonzero letters. This is the usual series-depth filtration.

The map induced by $t \mapsto (1-t)/(1+t)$ is

$$C(v_1 \cdots v_n) = \tau(v_n) \cdots \tau(v_1), \quad \begin{array}{c|ccccc} v & z & c & a & b & \bar{a} \\ \hline \tau v & c-b & z+b & b-\bar{a} & b & b-a. \end{array}$$

Here $\tau = -\phi^*$ incorporates the reversal of orientation. Every transformed admissible word is admissible, and change of variables gives $H(Cw) = H(w)$. Consequently the period map annihilates the graded, K -stable shuffle ideal

$$\mathcal{J} = \langle f - Cf : f \in \mathcal{A} \rangle_{\Pi}. \quad (5.2)$$

No restriction on weight, depth, or multiplier is made in this definition.

In the following polynomial, juxtaposition is concatenation and $\star$ is the usual series stuffle product, transferred to words by cumulative colors. Define

$$P_6 = 128588z^5a\bar{a} + 3138084z^5aa + 1592832z^4aza + 521184z^3az^2a - 216172z^6a + 48861((z^3a) \star (z^2c)) - 257176((z^5a) \star b), \quad T = \frac{P_6 - KP_6}{2}. \quad (5.3)$$

This is the exact target used by the earlier bounded certificate. It belongs to $F_2\mathcal{A}_7^-$. The elementary harmonic split gives $S_6 = g_{6,1} + \text{Im Li}_{6,1}(i, -1)$, while $\beta(7) = 61\pi^7/184320$. Using also $H(b) = -L$ gives

$$H(T) = 128588iR_6. \quad (5.4)$$

Both products in (5.3) use convergent inputs, so their stuffle expansion requires no regularization.

For precision, $\mathcal{R}_4 \subset F_4\mathcal{A}_7^-$ denotes the *same* inherited finite row family as in [12]. Its construction is retained in the delivered independent enumerator:

1. All weight-seven shuffle-minus-stuffle products with sum of input depths at most four. Inputs are admissible except for the permitted single factor $\text{Li}_1(1)$; the standard one-divergence comparison cancels its divergent term.
2. All convergent level-two-to-level-four distribution rows and their products with admissible complementary-weight words, provided the sum of depths is at most four. Products are expanded by stuffle.
3. The already proved imaginary S_2 and S_4 rows multiplied by real parts of all admissible complementary-weight words of depth at most two, again expanded by stuffle.

4. The inherited balanced Cayley rows and their specified products. In the adapted alphabet below, the seed u begins with a letter other than x , ends with a letter other than z , and satisfies $\min(n_z(u), n_x(u)) \geq 1$. A multiplier v is allowed when $|u| - \min(n_z(u), n_x(u)) + \text{dep}(v) \leq 4$ and $|u| + |v| = 7$. Its product is expanded by shuffle.

Imaginary projection, rational normalization, and duplicate removal give, in this order, 3460, 1394, 40, and 237 new nonzero rows, respectively: 5131 rows in total. These counts specify a formal enumeration, not dimensions of periods. The fourth family may be retained even though the new assertion explicitly imposes the complete ideal (5.2).

Theorem 5.1 (Full-Cayley extension of the weight-seven obstruction). *With the formal target and row family just specified,*

$$T \notin \mathcal{J}_7^- + \text{span}_{\mathbb{Q}} \mathcal{R}_4. \quad (5.5)$$

In particular the inherited finite proof search remains unsuccessful even after all convergent Cayley relations and arbitrary shuffle products of them have been included.

The distinction between the full ideal and its bounded intersection is essential. We verify a finite basis in $F_4 \mathcal{A}_7^-$, but the theorem allows $\mathcal{J}$ in the entire admissible word algebra. The next two subsections justify why those finite checks imply that stronger assertion.

5.2 Why the depth-preserving projector controls the entire ideal

We state and justify the precise prerequisite from [13], so the logic of the extension does not depend on a finite-rank extrapolation. Introduce

$$x = c - b, \quad y = a - \bar{a}, \quad h = a + \bar{a} - b, \quad \mathcal{E} = \{z, y, h, b, x\}.$$

Then

$$c = x + b, \quad a = (y + h + b)/2, \quad \bar{a} = (-y + h + b)/2.$$

In this basis admissibility means first letter different from x and last letter different from z . Depth is $|w| - n_z(w)$, conjugation acts by $(-1)^{n_y(w)}$, and C reverses a word, interchanges z, x , and multiplies by $(-1)^{n_h(w)}$.

Let $\tilde{\mathcal{A}}$ be the full shuffle algebra, including nonadmissible formal words. It is used here only algebraically. With the order $z < y < h < b < x$, every Lyndon word of length at least two is admissible. The shuffle-polynomial theorem [14] consequently gives

$$\tilde{\mathcal{A}} = \mathcal{A}[z, x].$$

Let R_0 be the algebra retraction setting the two *shuffle polynomial* variables z, x to zero. It preserves both letter counts, commutes with C, K , and is explicitly

$$R_0(w) = \sum_{u=0}^l \sum_{v=0}^r (-1)^{u+v} x^u \amalg w[u : |w| - v] \amalg z^v, \quad (5.6)$$

where l, r are the initial x run and terminal z run. Deleting an initial x and deleting a terminal z are commuting shuffle derivations; their finite Taylor substitutions prove (5.6). The factorials cancel because shuffle powers of a single letter are factorial multiples of its concatenation powers. In particular this is not literal erasure of letters inside a word, and assigns no analytic value to divergent integrals.

For nonempty w put

$$e(w) = \sum_{k=1}^{|w|} \frac{(-1)^{k-1}}{k} \sum_{w=w_1 \cdots w_k} w_1 \amalg \cdots \amalg w_k, \quad E = R_0 e, \quad (5.7)$$

where the blocks are consecutive and nonempty. The map e is the first Eulerian logarithm. It annihilates products of positive-degree factors and equals the identity modulo such products [15]. These facts also follow from the convolution logarithm of the identity character into the commutative shuffle algebra: its logarithm is an infinitesimal character, and the term $k = 1$ gives the congruence. Thus $e^2 = e$. Reversal permutes consecutive cuts, so e commutes with C ; it also commutes with K and preserves both counts.

The subspace $U = E(\mathcal{A})$ maps isomorphically onto the indecomposables of $\mathcal{A}$, and multiplication identifies $\mathcal{A}$ with $\text{Sym}(U)$. To see this directly, E kills positive-degree products and is the identity modulo them. Any homogeneous lift of an indecomposable basis therefore gives polynomial generators, by a triangular change from the Lyndon generators. The lifts are bigraded by (n_z, n_x) , and C exchanges these two degrees.

For a homogeneous $u \in U$ of degree (p, q) choose the orientation

$$Q_{\uparrow}u = \begin{cases} u, & p > q, \\ Cu, & p < q, \\ (u + Cu)/2, & p = q. \end{cases}$$

On each pair of different bidegrees this identifies the two transported generators. On the diagonal it retains the positive eigenspace of C and kills its negative eigenspace. Extending multiplicatively gives an algebra retraction whose kernel is exactly $\mathcal{J}$: on the quotient all $u - Cu$ vanish, hence C acts identically on every polynomial, and conversely $u - Cu$ are among the ideal generators. With $E_{\uparrow} = Q_{\uparrow}E$, the retraction is the finite formula

$$\Pi_{\uparrow}(w) = \sum_{k=1}^{|w|} \frac{1}{k!} \sum_{w=w_1 \dots w_k} E_{\uparrow}(w_1) \Pi \dots \Pi E_{\uparrow}(w_k), \quad \Pi_{\uparrow}(1) = 1. \quad (5.8)$$

Convolution exponentiation of (5.7), followed by R_0 and the multiplicative orientation, proves this expression. For completeness, $E(w) \in U$ even if a block w is not admissible: in degree at least two, the polynomial decomposition $\mathcal{A}[z, x]$ makes $w - R_0w$ a sum of positive-degree products, killed by e ; in degree one, R_0e kills the two extra generators directly.

The orientation never decreases the number of z 's in any nonzero output. Shuffle multiplication adds their counts. Consequently, on $\mathcal{A}$,

$$\Pi_{\uparrow}^2 = \Pi_{\uparrow}, \quad \ker \Pi_{\uparrow} = \mathcal{J}, \quad \Pi_{\uparrow}K = K\Pi_{\uparrow}, \quad \Pi_{\uparrow}(F_D\mathcal{A}) \subseteq F_D\mathcal{A}. \quad (5.9)$$

These are all-weight algebraic statements. Notice that $C\Pi_{\uparrow} = \Pi_{\uparrow}$ is neither required nor generally true.

The verifier evaluates (5.7) using a permutation formula; its equivalence with consecutive cuts is also elementary in every weight. For a permutation of the n positions let d be the number of descents of its inverse. A shuffle of consecutive blocks produces that permutation exactly when cuts are placed at those d required positions, with any additional cuts chosen freely. Its total coefficient in (5.7) is therefore

$$\sum_{k=d+1}^n \frac{(-1)^{k-1}}{k} \binom{n-1-d}{k-1-d} = (-1)^d \int_0^1 t^d (1-t)^{n-1-d} dt = \frac{(-1)^d}{n \binom{n-1}{d}}.$$

This is precisely the implemented coefficient. Repeated letters simply combine the integer multiplicities of equal output words. Thus the finite audits of the two formulas are convention checks; their equality at weight seven does not depend on extrapolating those audits.

Lemma 5.2 (The full ideal intersection has a finite spanning certificate). *If $\mathcal{B}_4$ is any rational basis of $F_4\mathcal{A}_7^-$, then*

$$\mathcal{J}_7^- \cap F_4\mathcal{A}_7 = \text{span}_{\mathbb{Q}}\{w - \Pi_{\uparrow}(w) : w \in \mathcal{B}_4\}.$$

Proof. Every displayed difference belongs to the full ideal by the kernel property and belongs to the bounded space by depth preservation. Conversely, for f in the intersection, $\Pi_{\uparrow}f = 0$, so $f = (1 - \Pi_{\uparrow})f$. Expanding f in the finite basis proves the opposite inclusion. No restriction on the construction of f inside $\mathcal{J}$ has been imposed. $\square$

5.3 The integer certificate and the proof

Use the adapted-letter encoding $Z = 0, Y = 1, H = 2, B = 3, X = 4$. Enumerate all length-seven words lexicographically, retaining exactly those with first letter different from X , last letter different from Z , at least three Z 's, and an odd number of Y 's. This gives a basis $\mathcal{B}_4$ of $F_4\mathcal{A}_7^-$ with 2546 elements. The exact-depth counts are 1, 35, 390, 2120. They can be checked directly from the enumeration, or by choosing the nonzero positions and projecting to odd Y parity.

The certificate's `separator.json` supplies an integer functional λ on this basis. It is obtained by expressing the inherited functional of [12] in adapted coordinates; the functional itself is not being claimed as a newly discovered vector. The new fact is the second row of the following exact certificate:

$$\lambda(R) = 0 \quad (R \in \mathcal{R}_4), \quad (5.10)$$

$$\lambda(w - \Pi_{\uparrow}w) = 0 \quad (w \in \mathcal{B}_4), \quad (5.11)$$

$$\lambda(T) = -\mathcal{N} \neq 0, \quad (5.12)$$

where the positive integer is

$$\begin{aligned} \mathcal{N} = & 186660627289236812620583691130496682 \cdot 10^{36} \\ & + 78843750753489297279472069700354208. \end{aligned}$$

The integer is an exact functional value, not a numerical residual of the polylogarithm conjecture. Its decimal expansion is also recorded in the certificate JSON.

Proof of Theorem 5.1. The standalone verifier regenerates the complete inherited row family, checks (5.10), and independently reconstructs each $\Pi_{\uparrow}(w)$ by (5.6)–(5.8). It verifies (5.11) for all 2546 basis words, and reconstructs (5.3) before checking (5.12). All arithmetic is integral or rational. Thus Lemma 5.2 shows that λ annihilates $\mathcal{J}_7^- \cap F_4\mathcal{A}_7^-$.

If, contrary to the theorem, $T = j + r$ with $j \in \mathcal{J}_7^-$ and $r \in \text{span } \mathcal{R}_4$, then $T, r \in F_4\mathcal{A}_7^-$ implies $j = T - r \in F_4\mathcal{A}_7^-$. Consequently $\lambda(j) = 0$; also $\lambda(r) = 0$ by (5.10). This contradicts $\lambda(T) \neq 0$ and proves (5.5). $\square$

For reproducibility, execute

```
python certificates/cayley/code/verify_complete_cayley.py
```

from the package root. It requires only Python's standard library. The verifier checks SHA-256 digests of the two retained source modules and the functional data before use. It regenerates 5131 rows, verifies 2546 complete-ideal basis differences, and checks the nonzero target pairing. There are 2300 nonzero projector differences; no claim of their linear independence is made. Additional finite audits compare the Eulerian permutation formula against consecutive cuts on all 780 words through weight four and check 2150 shuffle products, as well as retraction, conjugation, and depth conventions. These audits support the implementation; the preceding algebraic proof supplies the all-weight projector properties and the passage to the full ideal.

5.4 Consequences for further research

The immediate computational recommendation is sharper than in the inspected sources. Completing Cayley symmetry while retaining precisely $\mathcal{R}_4$ does not settle S_6 . A successful continuation must change the additional relation input: for example by admitting suitable higher-depth double-shuffle or distribution rows, by adding another functional equation, or by deriving a uniform generating identity that implies the target. A larger numerical-precision calculation does not address this missing algebraic information.

The theorem does not prove that every conceivable derivation must pass through depth five. It excludes exactly the full ideal plus the specified finite additional span; another depth-four family could carry information outside that span. Likewise, no conclusion for the revised S_8 candidate follows merely by analogy. Useful next questions are to identify the lowest-weight additional row whose projected class pairs nontrivially with λ , to test whether selected depth-five double-shuffle rows suffice, and to find a conceptual functional equation explaining the whole even-index family. These are exact proof-search questions, while $R_6 = 0$ itself remains the separate numerical-period conjecture.

6 Corrections, source audit, and integration safeguards

6.1 What was inspected and what was not established

The source inspection covered the canonical manuscript's status and relevant harmonic, finite-part, Stieltjes, Gamma, and Gaussian chapters; the incoming archives were extracted and their relevant theorems, normalizations, future questions, and verification notes compared. The most direct dependencies are the ordered-Hurwitz, triple-Stieltjes–Tornheim, coincident-Stieltjes, directional-zeta, and resonance-coordinate reports. Two additional pinned formal-algebra sources supply the finite separator and the depth-preserving Cayley projector. This is a focused mathematical audit of the material used here, not a line-by-line certification of the entire multi-volume manuscript or every incoming claim.

No false theorem was identified in the selected incoming cubic and coincident claims. In particular, the two previously supplied forms of the diagonal cubic Tornheim relation are compatible after the zeta functional equation is applied. Their common third-derivative coordinate does not disappear merely because a reflected cubic difference is evaluated in Section 3. The older corrections to the rejected S_8 coefficient vector and related proof-status warnings are already reflected in the inspected manuscript; they should not be presented as newly discovered errors.

There is one definite external formula correction, proved in the next subsection. The remaining items below are precise safeguards for integrating the present results. They are not allegations that the inspected source texts already make each prohibited inference.

6.2 A sign correction and its full logarithmic extension

For this comparison use the unshifted raw-power harmonic zeta function

$$\mathcal{H}_p(s) = \sum_{n \geq 1} \frac{H_n^p}{n^s}, \quad p \geq 0, \quad \mathcal{H}_0(s) = \zeta(s),$$

and define its regular Laurent coefficients by

$$\mathcal{H}_p(1+u) = \text{PP}_{u=0} \mathcal{H}_p(1+u) + \sum_{j \geq 0} \frac{(-1)^j}{j!} \tilde{\gamma}_{p,j} u^j. \quad (6.1)$$

Thus $\tilde{\gamma}_{0,j} = \gamma_j$, the ordinary Stieltjes constant, and $\tilde{\gamma}_{p,j}$ corresponds to Tin's ${}_p\tilde{\gamma}_j$ in equation (14) of [9].

Proposition 6.1 (Convergent identities for every logarithmic moment). *For all integers $p, j \geq 0$,*

$$\sum_{n \geq 1} \frac{H_n^p (H_n - \log n - \gamma) \log^j n}{n} = \tilde{\gamma}_{p+1,j} - \tilde{\gamma}_{p,j+1} - \gamma \tilde{\gamma}_{p,j}. \quad (6.2)$$

The series is an ordinary absolutely convergent series.

Proof. Since $H_n - \log n - \gamma = 1/(2n) + O(n^{-2})$, the Dirichlet series with numerator $H_n^p (H_n - \log n - \gamma)$ and every fixed logarithmic derivative converge normally in $\Re s > 0$. In $\Re s > 1$ its sum equals

$$\mathcal{H}_{p+1}(s) + \mathcal{H}'_p(s) - \gamma \mathcal{H}_p(s),$$

so every principal part cancels at $s = 1$. Comparing the coefficient of u^j in (6.1), and then multiplying by $(-1)^j j!$, proves (6.2). $\square$

With the convention (6.1), the first term on the right of equation (22) of version 1 of [9], printed page 5, must be $-_p \tilde{\gamma}_1$, in place of the printed $+_p \tilde{\gamma}_1$. This corrects that sign; no other term of that equation is being audited here. The correction agrees with the paper's own $p = 1$ equation (11). Indeed,

$$\tilde{\gamma}_{1,0} = \frac{\gamma^2 + \zeta(2)}{2}, \quad \tilde{\gamma}_{2,0} = \frac{\gamma^3 + 5\zeta(3)}{3}$$

give the two illustrative cases

$$\sum_{n \geq 1} \frac{H_n - \log n - \gamma}{n} = \frac{\zeta(2) - \gamma^2}{2} - \gamma_1, \quad (6.3)$$

$$\sum_{n \geq 1} \frac{H_n (H_n - \log n - \gamma)}{n} = \frac{5}{3} \zeta(3) - \frac{\gamma}{2} \zeta(2) - \frac{\gamma^3}{6} - \tilde{\gamma}_{1,1}. \quad (6.4)$$

The two displayed regular constants follow directly from $\sum_{n \leq N} H_n/n = (H_N^2 + H_N^{(2)})/2$ and $\sum_{n \leq N} H_n^2/n = H_N^3/3 + \sum_{n \leq N} H_n/n^2 - H_N^{(3)}/3$, together with $\sum_{n \geq 1} H_n/n^2 = 2\zeta(3)$.

6.3 Five safeguards with mathematical consequences

Retain the full resonant amplitude. In (3.2), for odd $N = p + r$, the counterterm contains the entire polynomial $P_p(u)/u$. Replacing it by its principal pole alone changes the constant and higher coefficients. The difference is observable already in

$$\text{FP} \int_0^1 \psi'(x) K(x) dx = 2\zeta'(2), \quad \text{FP} \int_0^1 \psi(x) K'(x) dx = -2\zeta'(2) - 2\zeta(2).$$

The harmonic correction in (3.4) is part of the identity, not a removable convention. A source implementation should store the completed germ, rather than store its pole residue and reconstruct a finite part heuristically.

A total derivative can have a nonzero finite-part integral. With the fixed endpoint coordinates,

$$\text{FP} \int_0^1 f'(x) dx = \text{CT}_{x \rightarrow 1} f(x) - \text{CT}_{x \rightarrow 0} f(x)$$

whenever the endpoint Laurent-log expansions exist. Thus $\text{FP} \int_0^1 (\psi K)' = -2\zeta(2)$, although the Gamma primitive in (3.14) has zero boundary constant difference. These two cases must be checked separately. A blanket rule that all regularized total derivatives integrate to zero would be false.

Coordinate cancellation is not arithmetic minimality. The locus $b = c = d$ in Section 2 is exactly the locus where the coefficients of the named additional coordinates in the displayed normal form vanish. The three-ray reconstruction is an invertible coordinate calculation. Neither statement proves that η, ρ, κ, T are arithmetically independent over a specified field of ordinary constants. Additional functional relations could alter a later minimal presentation. The proposed status language is “proved spanning normal form with an exact cancellation locus in that form.”

A finite part and a spectral derivative use different data. At $s = -m$, multiplication by $1/\Gamma(s)$ kills the holomorphic Mellin remainder in the constant Laurent coefficient of the raw-power series. It generally does not kill its contribution to the coefficient of $s + m$. Consequently, the polynomial formula for $C_{p,m}(a)$ cannot be differentiated with respect to the discrete label m to obtain a spectral derivative. Actual argument differentiation is valid and gives (4.30), including its residue term. This distinction is essential for any primitive or higher-Stieltjes extension.

The complete Cayley result still has a fixed additional family. The ideal $\mathcal{J}$ in Section 5 is unrestricted, but $\mathcal{R}_4$ is the enumerated additional family. The new theorem rules out every construction within $\mathcal{J}_7^- + \text{span}_{\mathbb{Q}} \mathcal{R}_4$. It does not rule out additional double-shuffle, distribution, or other functional information. Nor does it prove that the actual period residual is nonzero or that every possible proof must pass through depth five. The integer pairing is a formal algebraic certificate, not a lower bound for a period.

6.4 Proposed canonical status updates

The following replacements are suitable for the research-status record.

1. Mark the depth-four independent-slope request as answered by a spanning formula with convergent coordinates and a formal cancellation locus. Leave arithmetic minimality and a classification modulo all functional relations open.
2. Extend the distinct-seam reflected closure to arbitrary repeated seams by the completed common-seam transform and finite rational partial fractions. Keep the generic unreflected cubic reduction open.
3. Add raw harmonic powers with continuous outer shift as a separate family from elementary-symmetric harmonic numerators. Record the all-even centered cancellation and the finite-part polynomial theorem.
4. Replace the proposed “complete Cayley symmetry and retry the same bounded rows” step by the proved nonmembership theorem. Keep S_6 and the revised S_8 explicitly conjectural.
5. Add a version-specific note for Tin’s equation (22), with (6.2) as the proof of the corrected sign. Do not silently change the Laurent convention to conceal the discrepancy.

7 Further research questions and a programme of exact identities

These questions begin at the boundaries of the proved results. They ask for exact formulas, functional relations, or finite certificates; none is a conjectural assertion presented as a theorem. They are ordered roughly by proximity to computations already available in this package.

7.1 Ordered jets, derivatives, and antiderivatives

Q1. Differential closure of the depth-four coordinates. Equation (2.8) already removes T in favor of $\eta'(a)$, ordinary Hurwitz jets, and the convergent value $Z_2(2, 1; a)$. Derive analogous argument-derivative formulas for ρ and κ , then use stuffle and partial fraction identities to reduce the resulting depth-three and mixed-weight sums. A useful first result would be a finite differential system with all additive normalizations fixed by the shift laws. Determine the smallest *proved spanning* system before asking for arithmetic minimality.

Q2. Two or more blocks of equal slopes at arbitrary depth. The all-equal and one-distinguished-slope families have Gamma generators. At depth four, the fully independent formula identifies the first coefficients that such generators miss. Seek an exact two-block generator for slopes $(p, \dots, p, q, \dots, q)$, with the block lengths arbitrary. The target is a finite collection of lower-depth generating germs independent of the block lengths, or a proof that a proposed finite collection fails. The depth-four normal form supplies a mandatory first consistency specialization.

Q3. Depth five with an explicit coordinate ledger. At depth five the finite part needs the cubic jet of R_2 , the quadratic jet of R_3 , and the linear jet of R_4 . Compute the exact regularized stuffle constraints on these jets and construct an analogue of the three-direction reconstruction. The deliverable should include a convergent representative for every retained coordinate, its shift recurrence, and a symbolic certificate for each claimed cancellation. This is a finite linear problem after the analytic regular germs have been fixed; it is not automatically a period-independence problem.

Q4. Rational-shift character combinations. For conductors 2, 3, 4, 6, apply the exact root-of-unity filter to the convergent representatives of η, ρ, κ, T . Determine which character-weighted combinations eliminate the residual ordered coordinates and reduce to ordinary Hurwitz or Dirichlet L jets. The shift identities alone give translations, not distribution. A successful theorem must show the cancellation algebraically and retain every logarithmic scale term.

Q5. Values on a polar hyperplane. The transverse-ray formula assumes all prefix slopes are nonzero. For a family lying in one prefix hyperplane, compare three explicit prescriptions: take a finite part in a normal variable first, remove the ordered poles through R_d first, or approach along a specified two-parameter surface. Compute their exact differences at depth four and determine which prescriptions preserve a stated stuffle identity. No value is defined by substituting a zero denominator into the ray formula.

7.2 Coincident products and Gamma primitives

Q6. The first two-unreflected Stieltjes closure problem. Reflection evaluates $\text{FP} \int \psi^2 K^{2h+1}$ in this article, while one arbitrary Stieltjes derivative times a reflected polynomial always closes. Classify the combinations of

$$\text{FP} \int_0^1 \gamma_m^{(p)}(x) \gamma_n^{(q)}(x) K(x)^r dx$$

whose completed higher-depth coordinate cancels. Begin with $m + n \leq 2$ and total argument derivative order at most two. The output should be an exact kernel of a finite coordinate map, with a proof of each cancellation and fixed endpoint coordinates.

Q7. Evaluate the generic cubic Tornheim coordinate. The cubic difference (3.12) reduces a mixed moment without determining the common unreflected cubic value. Seek a functional relation for the diagonal Tornheim third derivative in the incoming cubic formula, or a normalized Barnes-type representation whose derivative can be proved to equal it. This must be a new analytic relation, not another rearrangement of an equivalent cubic identity. The existing Mellin representation is a viable starting point.

Q8. A reflected generating function beyond polynomials. Polynomial closure is finite because the cotangent derivative polynomials form a triangular basis. Determine a nonpolynomial family $P_\lambda(K)$ for which the completed integral has a holomorphic parameter generator and controlled coefficientwise endpoint completion. Trigonometric beta integrals suggest rational or complex-power families, but their changing endpoint exponents introduce new resonances. A useful theorem would identify the full local subtraction before any expansion in λ .

Q9. Multiple Gamma primitives with fixed constants. The log-Gamma moments here are first argument primitives of digamma moments. Construct analogous Barnes- G or higher-Gamma identities for repeated primitives, with the integration constant fixed at an explicit positive point and compatibility checked against the functional equation. The endpoint constant terms must be included at every integration by parts. A first target is a closed generator for the even cotangent moments at the next primitive level.

7.3 Harmonic powers and spectral regular coefficients

Q10. Compact all-degree formulas for centered values. The finite word recursion proves that every $F_p(-2r, 1/2)$ belongs to the ordinary multiple-zeta algebra. Seek a bivariate generator in p and r that improves on enumerating all compositions of p . Even a closed formula for the entire sequence $F_p(-2, 1/2)$ would extend the explicit table. State the exact MZV coordinate space used at each weight rather than assume every value reduces to products of ordinary zeta values.

Q11. Generalized harmonic products and their regular shifts. For a fixed polynomial in $H_n^{(r)}$, $r \geq 1$, the same Mellin calculus produces polynomial finite parts. Which numerator combinations make every nonpositive even principal part vanish at $a = 1/2$? The raw-power parity proof does not apply automatically when $H_n^{(r)}$ with $r > 1$ is present. Compute the finite Bernoulli conditions at successive orders and seek a necessary and sufficient generating criterion. This gives exact cancellation identities rather than inequalities.

Q12. Cancel the first global Mellin remainder. The first regular spectral coefficient at $s = -m$ contains global information absent from its finite part. Find finite combinations of harmonic-power series whose remainder integrals cancel by an exact kernel identity. Proposition 6.1 is a model: one differential combination has a normally convergent series and an explicit relation among higher harmonic Stieltjes constants. Generalize this cancellation to nonpositive centers and generalized harmonic numerators, with the normalization checked independently.

7.4 The Gaussian period candidates

Q13. Identify additional information detected by the separator. The inherited functional survives completion of the entire Cayley ideal. A useful next computation is to generate selected additional double-shuffle or distribution relations outside the frozen family, project them by $\Pi_\dagger$, and identify a row on which the functional is nonzero. Such a row is a

concrete source of information missing from the present search; it is not by itself a proof of S_6 . Continue until either an exact rational proof certificate for the target is obtained or a new fully specified separating certificate is proved. Higher-depth intermediate rows are permitted, but need not be assumed necessary a priori.

Q14. A functional bridge to S_6 and the revised S_8 . Seek a parameter-dependent polylogarithm identity, differential system with enough exact boundary data, or regulator relation whose specialization is the short candidate. Independently, repeat the precisely defined formal quotient analysis for the revised weight-nine target. The weight-seven certificate cannot be extrapolated to it. Any new numerical relation remains a candidate until such a functional or exact symbolic proof is supplied.

Q15. Formal verification of the finite-part and ideal mechanisms. The local completion lemma and the depth-preserving ideal-intersection lemma are reusable proof units. Formalize those units in a proof assistant, then import the finite rational certificate with an exact integer parser. For the analytic identities, formalize normal convergence after a finite local subtraction before importing coefficient formulas. This would turn the current reproducible computations into a smaller formally checked core without confusing floating-point diagnostics with theorem proofs.

8 Proof dependencies and reproducible verification

The all-index analytic statements are proved by finite local subtraction, normal convergence, and exact coefficient extraction. The formal ideal statement combines an all-weight algebraic argument with a finite exact rational certificate. The additional computations have two purposes: to detect normalization or transcription mistakes, and to make the finite algebraic part independently replayable. They do not replace the analytic arguments.

8.1 Exact calculations

The ordered verifier solves the regularized stuffle equations for the required jets, extracts all Laurent coefficients through the finite part, checks the diagonal Gamma specialization and shift identities, and reconstructs the three-direction matrix. It passes 23 exact symbolic assertions, including a control that rejects an intentionally changed coefficient of κ .

The harmonic symbolic verifier independently constructs the low-weight multiple-polylogarithm expansions from their differential equations and fixed endpoint constants. It checks the Bernoulli principal coefficients, the displayed finite parts, parameter transport, and selected higher centered cancellations. It passes 171 exact symbolic assertions. The all-index parity proof remains the reason that every centered even pole vanishes.

The reflected verifier checks 15 exact derivative-polynomial identities in addition to its numerical comparisons. The complete Cayley verifier regenerates all 5,131 inherited rows, verifies all 2,546 basis differences in the ideal intersection, and obtains the exact nonzero target pairing in Section 5. It also checks 780 independent Eulerian/convention cases and 2,150 shuffle products. Both retained upstream code modules are byte-for-byte copies of their pinned Git blobs. The verifier checks their SHA-256 hashes and the exact functional data before carrying out the calculation.

8.2 Independent numerical diagnostics

Family	Independent evaluation	Largest observed discrepancy
Ordered depth four	Original ordered sums with Euler–Maclaurin tails; four directional finite parts and four shift checks	$6.54 \cdot 10^{-34}$
Reflected moments	Central ordinary quadrature plus independent endpoint Laurent-log integration; 30 checks	$4.38 \cdot 10^{-47}$
Harmonic powers	Original Dirichlet head and digamma asymptotic tail; 72 Laurent coefficients including larger-cutoff replays	$3.41 \cdot 10^{-41}$

The ordered run uses 55 working decimal digits with 35 guard digits for its local coefficient calculation, and tests shifts $a = 1, 2$. The reflected run uses 50 decimal digits and includes higher Stieltjes/argument derivative pairs that test the nonconstant resonant amplitude. The harmonic run uses 58 decimal digits, shifts $1/2$ and $3/4$, and spectral centers $0, -1, -2$; its independent replay increases the finite head, tail order, and Cauchy sample count. The attached JSON files record all numerical settings and individual residuals.

These are floating-point diagnostics, not interval enclosures. The numbers in the last column are observed discrepancies, not rigorous error bounds. The finite symbolic checks are also not proof-assistant formalizations of the analytic theorems. Separate derivations and cross-reviews checked the ordered jet equations, the harmonic local table, their factorial conventions, and their shift identities.

8.3 Files and replay commands

The package includes the modular article sources, a self-contained `Exact_Resonant_Identities.tex`, the compiled PDF, exact certificate data, verification scripts and receipts, a source manifest, a claims ledger, and integration notes. The self-contained source inlines the section and bibliography inputs, so its compilation does not require network retrieval or access to the inspected repository.

The numerical and symbolic scripts use Python with `mpmath` and `sympy`; the complete Cayley verifier requires only the Python standard library. From the package root:

```
python3 build.py
python3 verification/verify_all.py --exact-only
python3 verification/verify_all.py
```

The last command includes the higher-index reflected checks and the full independent numerical runs. The detailed README lists separate commands so a reader can replay only the result under review. No search matrix, modular rank cache, numerical polylogarithm conjecture check, or external service is required for the exact Cayley proof.

The top-level checksums describe the delivered files. Verification receipts contain elapsed times and may change when replayed; source integrity checks for the actual certificate are independent of those timings. No numerical threshold is used to decide the nonzero exact integer pairing.

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